Risk Management and Financial Institutions

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2026-09-22

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Preface

Course Overview

This course provides a comprehensive introduction to risk management principles and their practical applications in banking and insurance sectors. Students will gain deep insights into how financial institutions identify, assess, and mitigate various types of risk while exploring the critical role risk management plays in ensuring organizational stability and success.

Through a blend of theoretical frameworks and real-world case studies, participants will develop the analytical skills and strategic thinking necessary to navigate the complex risk landscape of modern financial services.

Learning Outcomes

Upon successful completion of this course, students will be able to:

- Master fundamental risk management concepts, terminology, and theoretical frameworks
- Understand regulatory requirements and industry best practices in financial risk management
- Apply quantitative and qualitative methods to measure and evaluate risk profiles
- Compare and contrast various risk identification methodologies
- Evaluate existing risk management systems and processes within financial institutions
- Propose innovative risk mitigation strategies using financial market instruments
- Assess risk transfer mechanisms including derivatives, securitization, and reinsurance
- Integrate risk management considerations into broader business decision-making

Key Topics Covered

The course explores risk management from both banking and insurance perspectives, examining how these industries approach common challenges while addressing their unique risk exposures. Students will engage with current industry developments, emerging risks, and evolving regulatory landscapes that shape modern risk management practices.

References

The whole course is based on the following textbooks:

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.

1 Introduction to Risk Management

References

- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 6 - Risk Management (Subchapters 1-4)
- CFA Program Curriculum, Level I, Volume 6 - Portfolio Management and Ethical and Professional Standards
  - Reading 53 - Introduction to Risk Management

Learning Outcomes:

- Define risk and distinguish between financial risks and non-financial risks
- Apply the five-step risk management process to evaluate and prioritize organizational risks
- Compare the four risk modification strategies and select appropriate responses based on risk appetite and tolerance
- Describe the key components of risk governance including organizational structure, the three lines of defense, and the role of risk culture
- Explain how Enterprise Risk Management (ERM) integrates siloed risks into a holistic framework and apply stress testing to identify vulnerabilities

1.1 Foundational Concepts

1.1.1 What is Risk?

Risk is the exposure to uncertainty that can lead to variation in outcomes, particularly adverse outcomes that result in losses or failure to achieve objectives. Simply put: **risk is the possibility that bad things might happen**.

The key distinction to understand:

- **Uncertainty** means we don't know what will happen
- **Risk** means we can estimate the probabilities of different outcomes

When we talk about risk in practice, we're actually referring to three interconnected components:

1. **Risk Driver** - The underlying source of uncertainty (what could change?)
2. **Risk Position** - The amount exposed to that uncertainty (how much is at stake?)
3. **Risk Exposure** - The potential impact from that position (what could we gain or lose?)

Example

Consider a company holding ¥1,000,000 when news from Japan could move the currency by ±1%:

- The risk driver is the ±1% currency movement
- The risk position is ¥1,000,000
- The risk exposure is ±¥10,000 potential impact

1.1.2 Risk Management: Optimization, Not Elimination

Risk management is the process of defining acceptable risk levels, measuring current risk, and adjusting exposure to achieve organizational objectives. The critical insight: **risk management aims to optimize value, not minimize risk**.

Why? Because risk and return are inseparable. We cannot directly control returns—we can only manage risk to influence expected outcomes. This leads to the fundamental principle of finance: **there is no free lunch**. Higher returns require accepting higher risks.

Example

Consider these investment options:

- Government Bonds: ~2-3% annual return with minimal risk
- Corporate Bonds: ~5-7% return with moderate default risk
- Startup Equity: 20%+ potential returns with high probability of total loss

The goal isn't to avoid the startup investment—it's to understand the risk-return tradeoff and choose appropriately based on objectives and risk tolerance.

1.1.3 The Risk Landscape: Financial and Non-Financial Risks

Organizations face two broad categories of risk that frequently interact and amplify each other.

1.1.3.1 Financial Risks

Market Risk stems from changes in market prices and rates. Market risk includes equity price movements, interest rate changes, foreign exchange fluctuations, and commodity price volatility.

Credit Risk is the possibility that a counterparty won't meet their obligations. The challenge with credit risk is its rarity—defaults are infrequent events with limited historical data, requiring estimation through credit spreads, rating models, and financial ratio analysis.

Liquidity Risk manifests in two forms:

- *Funding liquidity risk*: inability to meet payment obligations
- *Market liquidity risk*: inability to exit positions without significant losses

The 2008 crisis demonstrated this dramatically when mortgage-backed securities became essentially unsellable at any reasonable price. The risk isn't just wider bid-ask spreads (a known cost) but the uncertainty of spreads widening unpredictably, especially during stress periods.

1.1.3.2 Non-Financial Risks

Operational Risk arises from failed processes, people, systems, or external events. These events are typically low-frequency but high-impact, making them difficult to predict and quantify.

Strategic Risk emerges from poor business decisions or flawed execution.

Reputation Risk can destroy value rapidly, especially in our social media age. It often results from other risk events and spreads faster than organizations can respond.

Model Risk occurs when we use wrong models or misapply correct ones. All models simplify reality—the danger comes when we forget these limitations. **Models fail in predictable ways.**

Tail Risk represents extreme "black swan" events that occur more frequently than normal distributions suggest. These outlier events in the probability distribution's tails can devastate unprepared organizations.

1.1.3.3 The Critical Distinction: Systematic vs. Idiosyncratic Risk

Systematic risk affects entire markets or economies—interest rate changes, recessions, geopolitical events. It cannot be diversified away.

Idiosyncratic risk is specific to individual companies or assets. While diversification can reduce it, the 2008 crisis showed how idiosyncratic risks can evolve into systematic ones when individual firm failures trigger system-wide contagion.

1.1.4 Risk Interactions: When 1+1 = 3

Risks rarely occur in isolation. They interact, amplify each other, and cascade through organizations in non-linear ways. The 2008 Financial Crisis provides a textbook cascade:

1. Housing prices decline (market risk)
2. Mortgage defaults surge (credit risk)

3. MBS securities become illiquid (liquidity risk)
4. Banks face funding crises (funding liquidity risk)
5. Counterparty concerns explode (credit risk amplification)
6. System-wide crisis emerges (systemic risk)

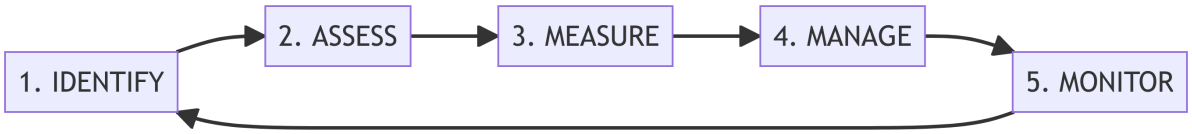
This cascade illustrates **wrong-way risk**—when exposure increases precisely as counterparty credit quality deteriorates. Banks buying credit protection from AIG, which was itself exposed to the same mortgage risks, exemplified this fatal correlation.

> **!** Important
>
> **The lesson:** risk models often fail to capture these interactions, making scenario planning and stress testing essential for identifying potential risk cascades before they occur.

1.2 The Risk Management Process

1.2.1 The Five-Step Risk Management Cycle

Risk management isn't a one-time exercise—it's a continuous cycle that integrates risk considerations into every organizational decision. The process flows through five interconnected steps, each building on the previous while feeding back into a continuous loop of improvement.



This cycle operates within a broader framework that provides structure and support:

- **Risk Governance** establishes the top-down direction, ensuring risk management aligns with organizational objectives.
- **Risk Infrastructure** provides the people, systems, and analytical tools needed to track and quantify risks.
- **Policies and Processes** translate high-level risk appetite into operational guidelines and limits.
- The **Three Lines of Defense** create organizational accountability—business operations own and manage risks, risk management provides oversight, and internal audit offers independent assurance.

> ❗ Important
>
> **A critical insight:** while organizations face countless potential risks, they're typically driven by a small number of key risk factors. Effective risk management focuses on identifying and managing these critical drivers.

1.2.2 Risk Identification—Finding What Could Hurt You

The first challenge is discovering what risks you face. This requires systematic exploration beyond the obvious, because the risks that destroy organizations are often the ones they didn't see coming.

Effective identification combines multiple approaches. **Risk workshops** bring together diverse perspectives to brainstorm potential threats. **Historical loss analysis** examines past failures—both internal and at peer organizations—to identify patterns. **Scenario analysis** asks "what could go wrong?" and follows the implications through to their logical conclusions. **Risk registers** provide systematic taxonomies of known risk types, while **external benchmarking** reveals risks other organizations have identified.

The challenge lies in navigating three categories of risk knowledge:

- **Known Knowns**: Risks we understand and can quantify (market volatility, credit defaults)
- **Known Unknowns**: Risks we're aware of but can't fully measure (cyber attack severity, pandemic impacts)
- **Unknown Unknowns**: "Black Swan" events we haven't imagined

The goal isn't to identify every conceivable risk—that's impossible. It's to cast a wide enough net to catch the risks that could materially impact objectives while remaining alert to signals of emerging unknown risks.

1.2.3 Risk Assessment—Prioritizing What Matters

Not all risks deserve equal attention. Assessment evaluates both the likelihood and potential impact of identified risks to focus resources where they matter most.

The classic risk matrix provides a simple but powerful prioritization tool:

Likelihood	Low Impact	Medium Impact	High Impact
High	Medium	High	Critical
Medium	Low	Medium	High
Low	Low	Low	Medium

High-likelihood, high-impact risks demand immediate attention. Low-likelihood, low-impact risks might be accepted as a cost of doing business. The challenging decisions involve the corners: high-impact but unlikely events (prepare contingency plans) and likely but low-impact events (automate controls).

Assessment isn't purely mechanical. It requires judgment about probability and impact, consideration of risk velocity (how quickly a risk materializes), and understanding of risk interconnections that could amplify impacts.

1.2.4 Risk Measurement—Quantifying the Unquantifiable

Measurement transforms vague concerns into actionable metrics. Different risk types require different measurement approaches, and choosing the right metric is crucial for effective management.

Foundation metrics apply across risk types:

- Probability distributions show the range of potential outcomes
- Standard deviation measures dispersion, though it assumes normal distributions that rarely exist in practice
- Value-at-Risk (VaR) estimates maximum likely loss at a confidence level
- Conditional VaR reveals average losses beyond the VaR threshold
- Stress testing explores performance under extreme scenarios

Specialized metrics target specific risks:

- For **market risk**, we use beta to measure systematic risk exposure, duration for interest rate sensitivity, and the "Greeks" for derivatives—delta (price sensitivity), gamma (delta's rate of change), vega (volatility sensitivity), theta (time decay), and rho (interest rate sensitivity).
- For **credit risk**, we estimate Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD). Credit ratings synthesize multiple factors into standardized risk grades.

Key Risk Indicators (KRIs) serve as early warning signals—metrics that change before risks materialize, providing time to respond.

The challenge in measurement is balancing precision with practicality. Perfect measurement is impossible, but approximate measurement that acknowledges uncertainty is far better than no measurement at all.

1.2.5 Risk Management—Choosing Your Response

Once risks are identified, assessed, and measured, organizations must decide how to respond. The four fundamental strategies can be combined for optimal results:

Risk Avoidance eliminates risk by not engaging in the activity. A company might avoid entering politically unstable markets or decline to develop products with uncertain liability. This provides certainty but foregoes potential opportunities. The key question: is what we're giving up worth the risk we're avoiding?

Risk Acceptance consciously retains risk, either because it's unavoidable or because the cost of mitigation exceeds potential losses. This takes two forms. **Self-insurance** means bearing losses as they occur or setting aside reserves. **Diversification** spreads risk across uncorrelated exposures, reducing concentration while accepting that some risk remains.

Risk Transfer shifts risk to parties better equipped to bear it. Insurance is the classic mechanism—insurers pool uncorrelated risks from many sources, using the law of large numbers to make aggregate losses predictable. The limitation: insurance becomes expensive or unavailable for risks that affect many parties simultaneously (pandemic, financial crisis).

Risk Shifting uses derivatives to alter risk distribution without transferring ownership. A company might use currency forwards to lock in exchange rates or buy options to limit downside while preserving upside. The choice between forwards (obligation) and options (right but not obligation) depends on whether you want certainty or flexibility.

The decision process starts with a fundamental question: is the risk within our appetite? If yes, we optimize management costs. If no, we must reduce exposure through some combination of the four strategies, considering costs, benefits, risk tolerance, and operational capabilities.

1.2.6 Risk Monitoring—Maintaining Vigilance

Risk management isn't "set and forget." Continuous monitoring ensures controls remain effective and strategies stay aligned with changing conditions.

Effective monitoring tracks KRIs against predetermined limits, providing early warning when risks approach unacceptable levels. It regularly updates risk assessments as business conditions change, tests control effectiveness through audits and reviews, runs stress scenarios to verify resilience, and adjusts strategies based on lessons learned.

Risk reporting transforms monitoring data into actionable intelligence. Effective reports share six characteristics:

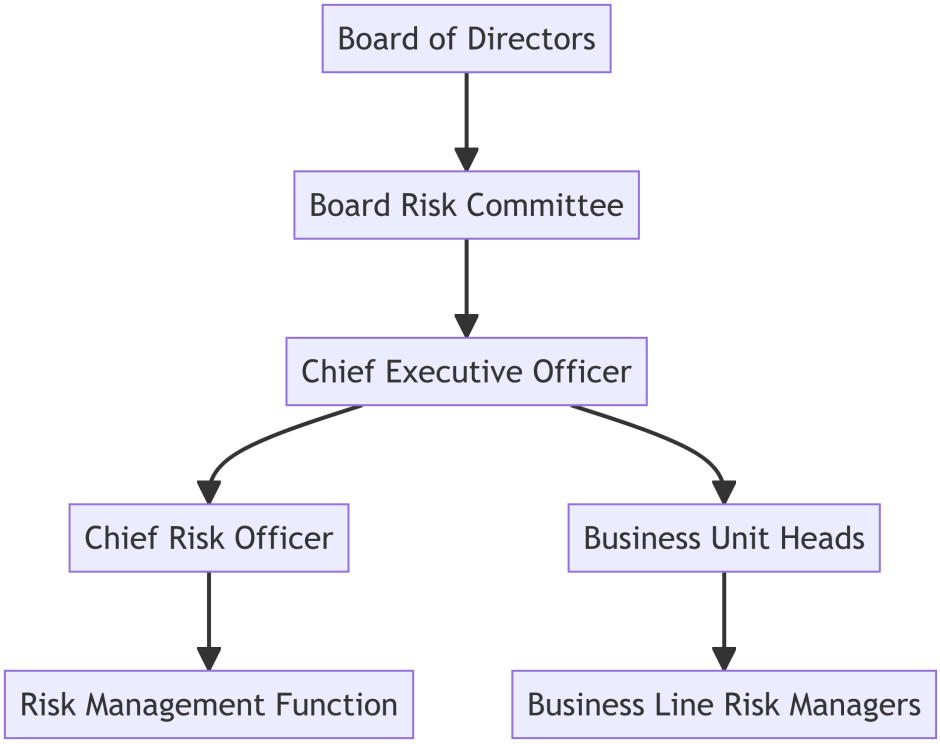
1. **Timely**—delivered frequently enough for decision-making, which varies by risk type (market risk might need daily reports, strategic risk quarterly)
2. **Accurate**—based on validated data and calculations stakeholders can trust
3. **Relevant**—focused on material risks for the specific audience, avoiding information overload
4. **Clear**—understandable by intended recipients, avoiding unnecessary jargon

5. **Actionable**—enabling informed decisions with clear implications and recommendations
6. **Comprehensive**—covering all material risk areas without gaps that could hide emerging problems

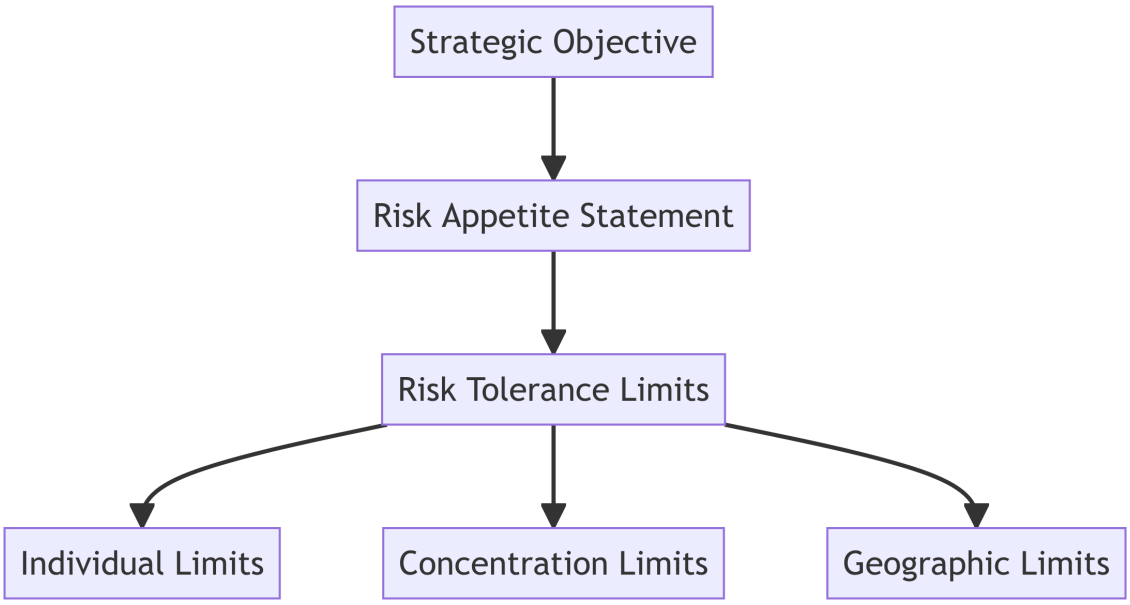
The monitoring phase feeds back into identification, creating a continuous cycle of improvement. New risks emerge, existing risks evolve, and yesterday's controls become today's vulnerabilities. Only through constant vigilance can organizations maintain resilience in an uncertain world.

1.3 Governance and Organization

Risk governance transforms risk management from compliance burden into strategic advantage through clear organizational structure and accountability. The hierarchy ensures risk awareness flows from boardroom to operations while information travels upward to inform decisions.



Risk appetite defines what risks an organization willingly accepts pursuing objectives, while **risk tolerance** sets acceptable variation boundaries. Organizations operationalize these through cascading limits:



Smart organizations set appetite well below capacity, maintaining buffers for unexpected shocks.

Risk budgeting transforms abstract risk appetite into concrete allocations across the organization. Like financial budgeting allocates scarce capital, risk budgeting allocates scarce risk capacity to its most productive uses.

Risk culture—shared attitudes and behaviors around risk—determines whether governance structures actually influence behavior. Strong culture exhibits four pillars: leadership tone from the top, clear accountability with consequences, effective challenge without retaliation, and risk-aligned incentives favoring long-term value over short-term profits. Poor culture reveals itself through profit-at-any-cost focus, suppressed concerns, and risk management viewed as impediment rather than enabler.

Modern risk governance **creates value by optimizing risk-return trade-offs**, enabling profitable activities others fear, and building competitive advantage through superior risk management. Success requires integration with business strategy, dynamic adaptation to changing conditions, and investment in capabilities matching strategic importance. The goal: risk management so embedded it becomes invisible yet omnipresent, influencing every decision without constraining innovation.

1.4 Modern Applications and Enterprise Risk Management

Enterprise Risk Management (ERM) represents a paradigm shift from isolated risk silos to **holistic risk management**. Traditional approaches created blind spots at intersections where risks interact and amplify—credit crises become liquidity crises become market crises. ERM acknowledges that organizations are driven by a small number of key risk factors cutting across traditional boundaries.

ERM rests on five interconnected pillars:

1. **Targets** align risk-taking with strategic objectives through enterprise-wide appetite statements and risk-adjusted metrics.
2. **Structure** creates accountability through board oversight and cross-functional committees.
3. **Identification and Metrics** establish common risk language and aggregate measures.

4. **Strategies** optimize risk-return trade-offs at portfolio level.
5. **Culture** embeds risk awareness as organizational mindset rather than isolated function.

Modern risk management looks forward through **stress testing** that explores extreme conditions. **Sensitivity analysis** changes single factors, **scenario analysis** combines multiple changes, and **reverse stress testing** identifies what could cause failure. Testing uses both historical scenarios (2008 crisis, COVID-19) and hypothetical events (cyber attacks, climate catastrophes) to reveal vulnerabilities that normal metrics miss.

New risk categories demand attention as the world evolves. **Climate risk** operates through physical damage, transition costs, and stranded assets. **Cyber risk** grows more sophisticated with cascading systemic impacts. **Artificial intelligence** creates accountability gaps and amplifies model risk. **Regulatory evolution** accelerates with dynamic requirements across multiple frameworks (Basel III, Solvency II, COSO).

! Important

The fundamental insight: ERM transforms risk from threat to opportunity, from cost to investment, from necessary evil to competitive advantage. Organizations that master ERM don't eliminate uncertainty—they harness it for sustainable value creation through intelligent risk-taking.

1.5 Practice Questions and Problems

Exercise 1: Risk Components

Identify the risk components:

- **Risk Driver**: What uncertain factor is creating the risk?
- **Risk Position**: What is your stake/exposure in this situation?
- **Risk Exposure**: What is the dollar amount potentially at stake?

i Scenario A: Currency Risk

You work for a German company that just signed a contract to receive $1,000,000 from a US client in 3 months. Today's exchange rate is €1 = $1.10, but it could move to anywhere between $1.00 and $1.20.

i Scenario B: Interest Rate Risk

Your company has a €5,000,000 loan with a variable interest rate, currently at 4%. The rate is reviewed every 6 months and could move ±2%.

Based on these two examples, how do the three risk components (driver, position, exposure) work together to create overall risk?

Exercise 2: Risk Identification

Review the real events from recent years provided below. For each event, complete the following:

- Identify all types of risks present in each event. Consider multiple perspectives - what could potentially go wrong for various stakeholders involved?
- Classify the identified risks into these four categories:
  - Financial/Market Risks
  - Operational/Process Risks
  - Human/Behavioral Risks
  - Technology/System Risks

This exercise helps develop pattern recognition skills for identifying and categorizing different risk types across real-world scenarios.

Event 1: Silicon Valley Bank Collapse (March 2023)

SVB focused heavily on tech startups and held many long-term government bonds. When interest rates rose, bond values fell. The bank announced losses, customers panicked, and $42 billion was withdrawn in one day via mobile banking.

Event 2: Ever Given Ship (March 2021)

A container ship got stuck in the Suez Canal for 6 days, blocking 12% of global trade. Oil prices rose, supply chains were disrupted, and some companies couldn't deliver products.

Event 3: GameStop Stock Surge (January 2021)

Reddit users coordinated to buy GameStop stock, driving the price from $20 to $347. Hedge funds that bet against the stock lost billions. Some trading apps restricted purchases.

Exercise 3: Risk Dominos

Pick ONE of the scenarios below and trace how problems might spread. Think about how one problem leads to the next:

- **Week 1:** Original event happens
- **Week 2:** What happens next?
- **Week 3:** What new problems emerge?
- **Week 4:** How does it spread further?
- **Week 5:** What's the final impact?

Critical Thinking Questions:

1. Which step was hardest to predict?
2. At what point could the company have stopped the cascade?
3. What does this tell you about managing risk?

Scenario A: Cyber Attack

A major bank's computer systems are hacked, and customer data is stolen.

Scenario B: CEO Scandal

The CEO of a major company is arrested for fraud.

Scenario C: Factory Fire

Your company's main manufacturing plant burns down.

Exercise 4: The Coffee Shop Dilemma

You're opening a coffee shop near campus. You've identified these potential problems:

- **Competitor opens next door:** High chance, medium impact
- **Coffee prices double:** Low chance, high impact
- **Students switch to energy drinks:** Medium chance, medium impact
- **Your espresso machine breaks:** High chance, low impact
- **Campus shuts down (like COVID):** Very low chance, very high impact

Suggest suitable solutions:

1. Rank these risks from 1-5 (1 = deal with first, 5 = worry about last)
2. For your top 3 risks, what would you do and why? For each one, you can:
    - **Avoid it** (don't do things that create the risk)
    - **Accept it** (live with it, maybe save money just in case)
    - **Transfer it** (get someone else to bear the risk - like insurance)
    - **Reduce it** (take action to make it less likely or less damaging)

Exercise 5: The Risk Management System

Imagine you're hired as "Chief Risk Officer" for a tech startup. The CEO says: "Just make sure nothing bad happens to us."

- What would you tell the CEO about their request? What's realistic and what isn't?

The CEO is more reasonable now. They want you to "manage risks systematically." Design your approach:

- What's the very first thing you'd do?
- What comes next?
- How would you measure things?
- What actions would you take?
- How would you keep watching?

Who should be responsible for risk management in the company?

- Just the risk manager
- Senior management
- All employees
- Board of directors
- External consultants

Exercise 6: Putting It All Together

Think about everything you've worked through today. In your own words, define:

- Risk
- Risk Management

Think of a major financial decision you've made (or will make soon) - buying a car, choosing a university, taking a job, etc.

- What risks did/do you face?
- How did/will you manage them?

You're in a job interview for a finance position. The interviewer asks: "Why should companies care about risk management?" Explain your opinion in 60s.

2 Revision and the 2007–08 Crisis

References

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 1 - Introduction: Risk-Return Trade-offs
  - Chapter 2 - Banks
  - Chapter 3 - Insurance Companies and Pension Plans
  - Chapter 4 - Fund Managers
  - Chapter 5 - Financial Instruments
  - Chapter 6 - The OTC Derivatives Market
  - Chapter 7 - Securitization and the Global Financial Crisis

Learning Outcomes:

- Identify the main risks of banks, insurers and funds, and distinguish capital from liquidity
- Explain the key mechanisms: diversification, beta, moral hazard, clearing and securitization
- Trace the 2007–08 crisis stage by stage: what went wrong, who could have stopped it, and what followed
- Apply the lessons of 2007–08 to a recent case (SVB, 2023)

2.1 Introduction

The questions below are for self-study: you should be able to explain each of them. If you missed the seminar, work through them with the textbook or an AI assistant, and verify any facts and figures you find.

Important

Remember: In risk management, questioning assumptions is essential. The most dangerous phrase remains: *"We've always done it this way!"*

2.2 Revision of Basic Concepts

1. What risks does a private investor face with a savings account, government bonds, equities, crypto, gold and a flat bought with a mortgage? How would you rank them on a risk matrix?
2. Why does combining two assets reduce risk even when their correlation is positive? Why does diversification weaken in a crisis?
3. Under CAPM, why is only systematic risk priced? How can a very volatile stock have a low expected return, and why can a negative-beta asset earn less than the risk-free rate?

4. Why must the average alpha across all investors be zero before costs? What does that imply for active funds?
5. What is the difference between bank capital and bank liquidity? Where does each appear on the balance sheet?
6. How can a bank fail without making a single bad loan? Why does leverage make banks fragile?
7. Define moral hazard and adverse selection, and give an example of each from banking and insurance.
8. How do life insurers, P&C insurers and CAT bonds manage risk? How can an insurer make a profit with a combined ratio above 100%?
9. Why can a closed-end fund trade below NAV? How do open-end funds and ETFs differ?
10. How does a "2 and 20" fee resemble an option, and what incentive does it create? Why do published hedge fund returns overstate what investors earn?
11. Why can a futures hedge create liquidity problems? What do CCPs and netting do, and does a CCP eliminate counterparty risk?

2.3 The 2007–08 Financial Crisis

1. Put the key events in order, from the peak in US house prices (2006) to TARP (October 2008). Were rising interest rates the cause?
2. What is the originate-to-distribute model, and why did it weaken lending standards? What role did refinancing play?
3. How do tranches work? How could AAA securities be built from subprime loans, and from BBB tranches in CDOs? What went wrong with ratings?
4. What did AIG Financial Products sell, why did it look like free money, and why did collateral calls bring AIG down?
5. Why did Northern Rock fail even though it met its capital requirements? What is wholesale funding, and why can a run bring down a solvent bank?
6. Why was Lehman allowed to fail while AIG was rescued two days later? What happened to money-market funds?
7. What did Basel III, Dodd-Frank and central clearing change, and what did they miss?
8. Pick one institution (Northern Rock, Bear Stearns, Lehman, AIG or SVB in 2023). What went wrong, who could have stopped it, what were the consequences, and what is the risk-management lesson?

3 Market Risk I: Value at Risk

References

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 11 - Value at Risk and Expected Shortfall
  - Chapter 12 - Historical Simulation and Extreme Value Theory
  - Chapter 13 - Model-Building Approach
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series. - Chapter 6 - Risk Management

Learning Outcomes:

- Calculate VaR using historical simulation and parametric methods for real portfolio data
- Identify when VaR violates subadditivity and explain the risk management implications
- Apply the square-root rule for time scaling and recognize when this assumption breaks
- Interpret VaR numbers in practical contexts and communicate limitations to stakeholders
- Evaluate why VaR fails to capture tail risk severity through real-world examples

3.1 Block A: What Risk Number Do We Need?

You've just been hired as a risk analyst at an equity trading desk. Your manager hands you a CSV file with the last 250 days of S&P 500 returns and says:

> The CEO will ask me tomorrow morning: 'How much could we lose on our $100 million SPY position in a day?' I need a single number by 4 PM.

Download Dataset (250 Days)

3.1.1 Activity 1: The Variance Approach

Load the SPY returns data (`SPY_returns_250.csv`) - one year of daily returns, small enough to inspect in Excel:

Task 1.1: Calculate the standard deviation of all 250 returns. What is the daily volatility?

Task 1.2: If we define risk as "2-sigma events," what loss level should we report for our $100M position?

Task 1.3: Find the best and worst daily returns in the dataset. Are they roughly symmetric around the mean? What does this tell you about using standard deviation for risk measurement?

P/L Convention

We use Profit/Loss (P/L) data where:

- Positive = profits (good)
- Negative = losses (bad)

For a $100M position:

- A -2% return means we lost $2 million
- A +1.5% return means we gained $1.5 million

This matches trader thinking and system reports.

3.1.2 Activity 2: Discovery Through Sorting

Task 2.1: Open `SPY_returns_250.csv` and sort the returns column from worst to best. What are the 5 worst daily returns?

Task 2.2: For 95% confidence with 250 observations:

- How many days would you allow as "exceptional bad days"?
- What loss value sits at this cutoff?
- For our $100M position, what dollar loss does this represent?

Task 2.3: Compare your answer from Task 2.2 with the "2-sigma" approach from Task 1.2. Which gives a larger loss estimate? Why might they differ?

3.1.3 Theory: Formalizing Value-at-Risk

VaR Definition

Value-at-Risk (VaR) is the loss threshold that will not be exceeded with $\alpha\%$ probability over a specified time horizon h.

Mathematical Definition:

For confidence level $\alpha \in (0, 1)$ and loss random variable L:

$$\text{VaR}_\alpha = \inf\{l \in \mathbb{R} : P(L > l) \leq 1 - \alpha\}$$

> VaR is the smallest loss value l such that the probability of exceeding l is at most $(1 - \alpha)$

Equivalently, using the loss distribution's quantile function:

$$\text{VaR}_\alpha = F_L^{-1}(1 - \alpha)$$

Where F_L^{-1} is the inverse cumulative distribution function (CDF), also called the quantile function.

> VaR is the $(1 - \alpha)$-quantile of the loss distribution

Three essential components:

1. Confidence level (α): The probability that losses will not exceed VaR
2. Time horizon (h): The period over which losses are measured
3. Loss amount: The monetary value at risk

Interpretation:

- "1-day 95% VaR = \$X" means we expect to lose less than \$X on 95% of days
- On 5% of days (about once per month), we expect to lose more than \$X.

3.1.4 Activity 3: Quick Application

Task 3.1: Interpretation Check:

- If your 1-day 99% VaR is \$4M, how often do you expect to exceed this loss?
- Does VaR tell us the maximum possible loss?
- Why might the CEO prefer one number (VaR) rather than the full distribution?

Task 3.2: Using your sorted SPY returns, calculate the 99% VaR for the \$100M position:

- Which observation should you use?
- What is the corresponding return?
- What is the VaR in dollars?

Task 3.3: Look at your 99% VaR calculation:

- Is this enough data to be confident about tail risk?
- What could go wrong with such limited tail observations?

3.2 Block B: Two Ways to Calculate VaR

Now we'll use 4 years of SPY data (`SPY_returns_1000.csv`) for a more robust analysis.

Download Dataset (1000 Days)

3.2.1 Theory: Historical Simulation vs. Parametric Approach

Let's calculate VaR using the historical simulation method.

Historical Simulation

Step 1: Sort all daily returns from worst to best
Step 2: Determine the position for your chosen confidence level α%:

- Number of exceptions allowed: $n \times (1 - \alpha)$
- Round conservatively to find the position k
- The k-th worst observation is your VaR

Step 3: Convert the return to dollar loss for your portfolio value

Now let's calculate VaR assuming returns follow a normal distribution.

Parametric Approach - Normal Distribution

Step 1: Calculate the distribution parameters from your data:

- Sample mean: $\mu = \frac{1}{n}\sum_{i=1}^{n} r_i$
- Sample standard deviation: $\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(r_i - \mu)^2}$

Step 2: Apply the parametric VaR formula using the appropriate z-score:

$$\text{VaR}_\alpha = -(\mu + \sigma \times z_{1-\alpha})$$

where $z_{1-\alpha}$ is the standard normal quantile
To find $z_{1-\alpha}$, you can use the Excel function `NORM.INV(1- , 0, 1)`

3.2.2 Activity 4: When Methods Diverge

Let's systematically compare the two main VaR approaches.

Task 4.1: Create a comparison table:

- Confidence levels = 90%, 95%, 97.5%, 99%, 99.9%
- Calculate VaR using historical simulation and parametric approach

Task 4.2: Discussion Questions

1. At which confidence levels do the methods diverge most?
2. Does the parametric 99.9% VaR exceed the worst historical loss? What does this imply?
3. Which method would you trust more for extreme events?

3.2.3 Theory: Time Scaling and the Square-Root Rule

Scaling VaR Across Time Horizons

The Square-Root Rule: Under specific assumptions, VaR scales with the square root of time:

$$\text{VaR}_{h\text{-day}} = \text{VaR}_{1\text{-day}} \times \sqrt{h}$$

Required assumptions:

1. Returns are independent across days (no autocorrelation)
2. Returns are identically distributed (stable volatility)
3. No mean reversion or momentum

3.2.4 Activity 5: Square-Root Rule

Task 5.1: Test the square-root rule with real data

- Calculate 10-day returns
- Calculate actual 10-day 99% VaR

- Calculate 10-day 99% VaR using the square-root rule from daily returns
- When might this scaling break down?

Task 5.2: A risk manager says:

> Our 1-day VaR is \$5M, so our annual VaR is $\$5M \times \sqrt{250} = \$79M$.

- What's wrong with this logic?

> **Key Insight**
>
> The square-root rule is widely used but fragile. In crisis periods when you need VaR most, the independence assumption fails spectacularly as correlations spike and volatility clusters.

3.2.5 Activity 6: Discussion

Task 6.1: You now have several VaR numbers from different methods. How do you decide which to report to the CEO?

Task 6.2: What key information is still missing from any VaR number?

3.3 Block C: When VaR Fails

3.3.1 Activity 7: The Diversification Paradox

Consider two corporate bonds in your portfolio:

	Bond A	Bond B
Notional	\$100M	\$100M
Default Probability	0.6%	0.6%
Recovery if Default	\$0	\$0
Mutual Correlation	0 (independent)	0 (independent)

Task 7.1: Calculate 99% VaR for Bonds A and B separately

- In 99% of scenarios, what happens?
- In 1% of scenarios, what is the loss?
- What is the 99% VaR value?

Task 7.2: Calculate 99% VaR for the combined portfolio (A + B)

- Probability that neither defaults?
- Probability that at least one defaults?
- Probability that both default?
- What is the 99% VaR value?

Task 7.3: Compare your results

Task 7.4: Your bank could split the portfolio into two separate legal entities, each holding one bond. What happens to regulatory capital requirements if capital is based on VaR?

Task 7.5: A trader discovers that selling deep out-of-the-money options shows zero VaR most days. Should you be concerned? Why?

Task 7.6: You're allocating risk limits to two trading desks. If Desk A has VaR limit of \$10M and Desk B has \$8M, can you be sure the combined risk is less than \$18M?

Real-world consideration

This property contributed to the 2008 crisis. Banks created structures that minimized VaR while concentrating tail risk. When the "impossible" happened, multiple firms failed simultaneously.

3.3.2 Theory: Coherent Risk Measure

Coherent Risk Measure Axioms

A risk measure R is called **coherent** if it satisfies four axioms:

1. **Subadditivity** (Diversification should reduce risk)

$$R(L_1 + L_2) \leq R(L_1) + R(L_2)$$

2. **Positive Homogeneity** (Scaling positions scales risk proportionally)

$$R(\lambda L) = \lambda R(L), \text{ for every } \lambda > 0$$

3. **Monotonicity** (Bigger losses mean bigger risk)

$$R(L_1) \leq R(L_2) \text{ if } L_1 \leq L_2$$

4. **Translation Invariance** (Adding cash reduces risk by that amount)

$$R(L + a) = R(L) - a$$

VaR violation: VaR can fail subadditivity, especially for:

- Portfolios with fat-tailed distributions
- Concentrated credit exposures
- Non-linear derivatives positions

Key insight: VaR satisfies only 3 of the 4 axioms. This single failure means VaR is not "coherent" in the mathematical sense.

3.4 Block D: Real Applications and Limitations

3.4.1 Activity 8: Reflection Questions

Task 8.1: If VaR can show that diversification increases risk, should we still use it?

Task 8.2: How might traders "game" a VaR-based risk limit system?

Task 8.3: What additional information would you want beyond VaR?

3.4.2 Case Study: JP Morgan's London Whale (2012)

JP Morgan's $6.2 billion loss illustrates how VaR model changes can mask growing risks.

Late 2011 - Original Model:

- Historical simulation VaR using full dataset
- Result: VaR = $95-132 million
- Status: Approaching risk limits
- CIO position size: ~$157 billion notional

January 2012 - New "Enhanced" Model:

- Switched to parametric (Gaussian) VaR approach
- Used shorter, calmer data window
- "Simplified" correlation calculations
- Result: VaR dropped to $54-67 million overnight
- Status: Comfortably within limits again

The Key Changes:

Aspect	Old Model	New Model	Impact
Method	Historical simulation	Parametric/Gaussian	Assumes normality
Data period	Multiple years	Shorter, recent only	Excluded crisis volatility
Weighting	Time-weighted	Equal-weighted	Reduced recent volatility impact
Implementation	Established system	Excel spreadsheet	Introduced formula errors

What Actually Happened:

- **The Excel error:** Divided by sum instead of average of volatilities (understated risk by ~50%)
- **VaR breaches:** The desk exceeded its VaR limit on 71 days in Q1 2012
- **Actual losses:** Daily losses exceeded $200M multiple times (3x the reported VaR)
- **Model reversion:** In April 2012, switched back to old model → VaR jumped to $129M

The Governance Failures:

- Model change approved without proper validation
- Risk committee not informed of methodology change
- Breaches explained away as "technical" or "model issues"
- Traders influenced risk measurement process

3.5 Key Takeaways

- **VaR is simple and intuitive** but has critical limitations
- **Different stakeholders** need different VaR parameters
- **VaR fails to capture**:
  - Tail risk severity
  - Liquidity risk
  - Model risk
  - Correlation breakdowns
- **Real failures** (London Whale, 2008 crisis) show VaR's limitations

4 Market Risk II: Expected Shortfall

References

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 11 - Value at Risk and Expected Shortfall
  - Chapter 12 - Historical Simulation and Extreme Value Theory
  - Chapter 13 - Model-Building Approach
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series. - Chapter 6 - Risk Management

Learning Outcomes:

- Calculate Expected Shortfall using historical simulation, parametric, and Monte Carlo methods
- Demonstrate why ES satisfies coherence properties (subadditivity) and explain the risk management implications
- Generate and analyze Monte Carlo simulations with different distributions to assess tail risk
- Compare Basel II (VaR-based) and Basel III (ES-based) capital requirements and explain why regulators made this shift
- Evaluate real-world risk model failures through the UBS case study and identify when VaR dangerously underestimates tail risk

4.1 Block A: Beyond VaR - What Happens in the Tail?

Last week, you reported to your CEO that the 1-day 99% VaR for your $100M SPY position was $4.2M. This morning, she calls:

> We just lost $8.7 million yesterday! You said we'd only exceed $4.2M once in 100 days. Should I fire you, or is this normal?

4.1.1 Activity 1: The VaR Breach Investigation

Open your `SPY_returns_1000.csv` file from last week.

Download Dataset (1000 Days)

Task 1.1: Find all daily returns worse than your 99% VaR threshold

- How many observations are there?
- List the worst return in your dataset?
- What's the average loss on days when VaR is breached?

Task 1.2: Your colleague says:

> Don't worry, VaR breaches are rare - only 1% of the time! The actual loss doesn't matter.

- What critical information is your colleague missing?

Task 1.3: Two portfolios both have 1-day 99% VaR of $10M:

- Portfolio A: Government bonds
- Portfolio B: Written put options on SPY
- On a day when both breach their VaR, which likely loses more? Why?

4.1.2 Activity 2: Tail Risk Discovery

Let's examine what happens "beyond VaR" using your SPY data.

Task 2.1: For the worst 5% of days:

- Calculate the minimum loss
- Calculate the maximum loss
- Calculate the average loss
- How does the average compare to your 95% VaR?

Task 2.2: Risk Reporting Dilemma

You need to report risk to three stakeholders. Using your calculations:

Stakeholder	Their Question	VaR Answer	What's Missing?
CEO	"What's our typical bad day loss?"	$4.2M	?
CFO	"If things go bad, how much cash do I need?"	$4.2M	?
Regulator	"Can you survive a tail event?"	$4.2M	?

Task 2.3: VaR Breach Scenarios

Your risk report for last year shows:

	Bank A	Bank B
99% VaR	$10M	$10M
Breach 1	-$11M	-$10.5M
Breach 2	-$12M	-$11M
Breach 3	-$10.5M	-$45M

- Is it acceptable for 99% confidence?
- Which bank had worse tail risk?
- What does this tell you about VaR as a risk measure?

> ⚠ Key Insight
>
> VaR tells you the **frequency** of bad events and the **threshold** but nothing about the **severity** when that threshold is breached.

In crises, what kills firms isn't the frequency of bad days—it's the magnitude of losses on those days.

4.1.3 Theory: Conditional Expectation in the Tail

Expected Shortfall Definition

Expected Shortfall (ES) measures the average loss during bad days when things go worse than your VaR threshold.
Also called Conditional VaR (CVaR) or Expected Tail Loss (ETL), ES answers a critical question: When we have a really bad day that exceeds our VaR limit, how bad should we expect it to be?
ES is always worse than (or equal to) VaR because it's the average of all the losses in the tail, including the very worst outcomes.

Mathematically, for confidence level α:

$$ES_\alpha = E[L \mid L \geq VaR_\alpha]$$

This is the **average of all losses in the worst** $(1-\alpha)$ **portion of the distribution**, including the VaR observation itself.

More precisely for continuous distributions:

$$ES_\alpha = \frac{1}{1-\alpha} \int_{VaR_\alpha}^{\infty} L \cdot f(L)\, dL$$

For discrete historical data with n observations:

$$ES_\alpha = \frac{1}{k} \sum_{i=1}^{k} L_i$$

where:

- $k = \lceil n(1-\alpha) \rceil$ is the number of worst losses to average
- $L_1 \geq L_2 \geq ... \geq L_k$ are losses sorted from worst to best
- $L_k = VaR_\alpha$ (the k-th worst loss is the VaR)

4.2 Block B: Expected Shortfall - Measuring Tail Risk

4.2.1 Theory: Calculating Expected Shortfall

Historical Simulation Method for ES

Step 1: Sort all daily returns from worst to best
Step 2: Determine how many tail observations to include:

$$k = \lceil n \times (1-\alpha) \rceil$$

Step 3: Calculate the average of these k worst observations:

$$ES_\alpha = \frac{1}{k}\sum_{i=1}^{k} L_i$$

Step 4: Convert to dollar loss for your portfolio value

Parametric Approach for ES (Normal Distribution)

For normally distributed returns:
Step 1: Calculate distribution parameters from your data:

- Sample mean: μ
- Sample standard deviation: σ

Step 2: Apply the ES formula for normal distribution:

$$ES_\alpha = -(\mu - \sigma \times \frac{\phi(z_\alpha)}{1-\alpha})$$

where:

- z_α is the standard normal quantile at confidence α
- $\phi(z_\alpha)$ is the standard normal PDF evaluated at z_α

The PDF can be calculated as:

$$\phi(z_\alpha) = \frac{e^{-z_\alpha^2/2}}{\sqrt{2\pi}}$$

Excel implementation:

- For PDF: `NORM.DIST(z_ , 0, 1, FALSE)`
- For z-score: `NORM.INV(1- , 0, 1)`
- Complete ES formula: `=-1*(mean - stdev*NORM.DIST(NORM.INV(1-,0,1),0,1,FALSE)/(1- ))`

Note: The negative sign converts P/L returns (where losses are negative) to positive loss values, matching our VaR convention.
Step 3: Convert to dollar loss for your portfolio value

4.2.2 Activity 3: Comparing ES Methods

Using your `SPY_returns_1000.csv` data, let's compare ES calculation methods.

Task 3.1: Create a comprehensive risk comparison table with the following structure:

- Confidence levels = 90%, 95%, 97.5%, 99%, 99.9%
- Calculate ES using historical simulation and parametric approach
- ES/VaR ratio for each method

Task 3.2: Discussion Questions

1. At which confidence levels do the methods diverge most?

2. How does the ES/VaR ratio change as confidence level increases?
3. For 99.9% confidence, does parametric ES exceed the worst historical loss?
4. Which method would you recommend for setting risk limits?

4.2.3 Activity 4: The Coherence Property - Subadditivity

Let's verify that ES satisfies the subadditivity property that VaR violates.

Consider two corporate bonds in your portfolio:

	Bond A	Bond B
Notional	\$100M	\$100M
Default Probability	0.6%	0.6%
Recovery if Default	\$0	\$0
Mutual Correlation	0 (independent)	0 (independent)

Task 4.1: Calculate 99% ES for Bonds A and B separately

- In 99% of scenarios, what happens?
- In 1% of scenarios, what is the loss?
- What is the 99% ES value?

Task 4.2: Calculate 99% ES for the combined portfolio (A + B)

- Probability that neither defaults?
- Probability that at least one defaults?
- Probability that both default?
- What is the 99% ES value?

Task 4.3: Compare your results

> **Real-world Implications**
>
> The subadditivity property matters because:
>
> - It ensures diversification is always rewarded
> - Risk limits can be allocated consistently across desks
> - Portfolio risk is never understated when aggregating positions
> - Capital requirements properly reflect risk reduction from diversification

4.2.4 Theory: Regulatory Context - From VaR to ES

> **Basel III's Shift to Expected Shortfall**
>
> **The regulatory evolution:**
>
> - **Basel II (until 2016):** Market Risk Capital based on VaR(99%, 10-day) with multiplier of 3
> - **Basel III (2016-present):** Market Risk Capital based on ES(97.5%, stressed) with multiplier of 1.5

Key change: After the 2008 crisis, regulators switched from VaR to ES because:

- VaR missed tail severity (banks failed despite "acceptable" VaR)
- ES is mathematically coherent (rewards diversification)
- ES is harder to manipulate

Stressed calibration requirement:

Basel III requires banks to calculate ES using both:

- **Recent period:** Last 250 trading days
- **Stressed period:** The historical 250-day window that produces highest ES for current portfolio

$$\text{Capital} = 1.5 \times \max(ES_{recent}, ES_{stressed})$$

The stressed period is found by testing rolling 250-day windows across historical data (typically capturing crisis periods like 2008 or March 2020).

Note: We'll explore the full regulatory framework in detail in a later seminar. For now, these exercises demonstrate why regulators made this shift.

4.2.5 Activity 5: Practical Applications

Task 5.1: In 2016, Basel III shifted from VaR to ES for market risk capital. Using your SPY data, compare the old and new requirements:

- Calculate 99% VaR (old Basel II requirement)
- Calculate 97.5% ES (new Basel III requirement)
- Despite the lower confidence level, is the new requirement more or less conservative?

Task 5.2: Calculate 99% VaR and 97.5% ES for all 250-day rolling windows.

- Which 250-day window produces the highest VaR and ES?
- Compare the results with the most recent 250-day window.
- If you had to hold capital for the worst-case historical period, how much more would you need?

4.3 Block C: Monte Carlo Simulation for Risk Measurement

4.3.1 Theory: From Historical Data to Simulated Futures

Monte Carlo Methods in Risk Management

Why simulate?

- Historical data is limited (only one path of history)
- Need to stress test scenarios that haven't happened yet
- Can incorporate complex dependencies and dynamics

Basic Monte Carlo for VaR/ES:

1. Specify a distribution for returns (normal, student-t, etc.)

2. Generate thousands of possible scenarios
3. Calculate portfolio value for each scenario
4. Extract VaR and ES from simulated distribution

Industry practice uses more sophisticated models:

- **Stochastic Volatility:** Volatility itself is random (GARCH, Heston)
- **Jump Diffusion:** Sudden jumps in addition to normal movements
- **Regime Switching:** Different distributions in calm vs stressed periods
- **Copulas:** Complex dependency structures between assets

4.3.2 Activity 6: Basic Monte Carlo in Excel

Task 6.1: Create 5000 simulated daily returns for your SPY position:

- In Excel: `=NORM.INV(RAND(), mean, stdev)`
- Use mean and stdev from your historical data
- Calculae 95%/99% VaR and ES using historical method

Task 6.2: Real returns have fatter tails than normal. Let's use Student-t distribution and generate another 5000 simlated daily returns

- `=T.INV(RAND(), degrees_of_freedom) * stdev + mean`
- Use 5 degrees of freedom for fat tails
- Calculae 95%/99% VaR and ES using historical method

Task 6.3: Compare results from historical simulation (actual), parametric method (theoretical), and Monte Carlo simulations. Which methods give similar results? Why might they differ?

4.3.3 Activity 7: Convergence and Stability

Task 7.1: Using Student-t distribution, calculate 97.5% ES with different numbers of scenarios:

- E.g., 200, 600, 1000, 2000, 5000, 10000 scenarios
- Plot ES estimate vs number of scenarios. At what point does it stabilize?

Task 7.2: Run your 1000-scenario simulation 5 times (press `F9` to recalculate) and write down the results for 99% VaR and 97.5% ES. How stable are your estimates? What does this imply for risk management?

Real-World Implementation

Banks typically use:

- 10,000+ scenarios minimum
- Multiple risk factors (not just single asset)
- Historical simulation with bootstrapping
- Filtered Historical Simulation (combining historical data with volatility models)

The trade-off: More scenarios = more accuracy but slower calculation. For intraday risk, banks often use fewer scenarios for speed.

4.4 Block D: Case Study - UBS and the 2007-2008 Financial Crisis

UBS, Switzerland's largest bank, suffered the biggest losses of any European bank during the 2007-2008 financial crisis. Despite sophisticated risk models and AAA-rated assets, the bank lost over $37 billion on mortgage-related securities. This case illustrates how VaR models can catastrophically fail and why Expected Shortfall matters.

Before the Crisis (2005-2007):

- UBS held $50+ billion in "Super Senior" CDO tranches (the safest part of mortgage securities)
- These were rated AAA - supposedly safer than US government bonds
- Risk models (like VaR) showed minimal danger because:
  - Historical data (2003-2006) showed tiny default rates
  - Mortgages across different regions were assumed independent
  - The bank bought insurance from monoline insurers as protection
  - Assumed normal distributions and stable correlations

During the Crisis (2007-2008):

- October 2007: UBS announces $3.4 billion loss (first major bank to report)
- Full year 2007: $18.7 billion in writedowns
- Q1 2008: Another $19 billion in losses
- Stock price fell 70%, CEO and entire senior management resigned

Why Everything Failed:

- US housing collapsed nationwide (not regionally as models assumed)
- AAA-rated securities became worthless
- Insurance companies failed (they insured the same mortgages)
- Market liquidity vanished - couldn't sell at any price
- Losses exceeded any historical scenario by 10-20x

4.4.1 Activity 8: Discussion Questions

Task 8.1: Model Assumptions

- If historical data shows no defaults in AAA securities, what should your VaR be?
- Should risk models include events that have never happened?
- How do you model risk for new products with limited history?

Task 8.2: Risk Aggregation

- UBS claimed positions were "diversified" across different mortgages. What went wrong?
- If correlation changes from 0.2 to 0.9 in crisis, how does this affect portfolio risk?
- Can diversification make risk disappear, or just hide it?

Task 8.3: Lessons Learned

- Would higher capital requirements based on ES have prevented these losses?
- If everyone uses similar risk models, what systemic risks emerge?
- Why might a model that works for 99% of days still destroy a bank?

⚠ The Key Lesson

VaR's blind spot: It tells you the threshold you won't exceed 99% of the time, but nothing about what happens in that 1%.
UBS's fate: The 1% tail event wasn't just slightly worse than VaR - it was catastrophically worse. This is why regulators switched to Expected Shortfall after the crisis.
Remember: Risk models work well in normal times. They fail precisely when you need them most - during crises when correlations spike, liquidity vanishes, and "impossible" events happen.

4.5 Key Takeaways

- **VaR's fatal flaw:** It tells you nothing about severity when breached—ES measures what actually happens in the tail
- **Coherence matters:** ES always rewards diversification (subadditive), while VaR can punish it, leading to perverse incentives
- **Distribution choice is critical:** Normal distributions underestimate tail risk; fat-tailed distributions (Student-t) better capture extreme events
- **Monte Carlo extends our toolkit:** When historical data is limited, simulation helps explore scenarios that haven't happened yet
- **Regulatory evolution has a reason:** Basel III's shift to ES wasn't arbitrary—it directly addressed failures revealed in 2008
- **Models fail when you need them most:** UBS lost $37B despite "safe" VaR—risk models work in normal times but break in crises when correlations spike and liquidity vanishes
- **The practical bottom line:** Always ask "If things go bad, HOW bad?"—that's ES, and it might save your firm

5 Market Risk III: Model Validation and Regulatory Capital

References

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 11 - Value at Risk and Expected Shortfall
  - Chapter 12 - Historical Simulation and Extreme Value Theory
  - Chapter 13 - Model-Building Approach
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series. - Chapter 6 - Risk Management

Learning Outcomes:

- Design and implement backtesting procedures for VaR models using the traffic light approach and statistical tests
- Evaluate the challenges of backtesting Expected Shortfall and apply practical solutions
- Calculate stressed VaR and stressed ES measures using historical crisis periods and explain their role in regulatory capital
- Compare Basel II, Basel 2.5, and Basel III market risk frameworks and analyze the impact on capital requirements

5.1 Block A: Backtesting VaR Models

5.1.1 Activity 1: Testing Your VaR Model

Your risk committee asks:

> Are the VaR models we built last week actually working?

A 99% VaR is a promise: the loss should exceed it on only 1% of days. To check the promise fairly, we estimate VaR on one period and test it on a later period that the model has never seen (a train-test split).

Using your `SPY_returns_1000.csv` file:

Download Dataset (1000 Days)

- **Training period:** Days 501-750 (250 days) - calculate VaR on this period
- **Testing period:** Days 751-1000 (250 days) - backtest your model on this period

Task 1.1: Using only days 501-750 (training set):

- Calculate 95% and 99% VaR using historical simulation (the k-th worst return, as in Week 3)

- Calculate the 99% historical VaR a second way, with Excel's percentile function
- Calculate 95% and 99% VaR using the parametric approach

Task 1.2: Apply your VaR thresholds to the test period (days 751-1000):

- Count the days on which the loss exceeded your 95% VaR
- Count the days on which the loss exceeded your 99% VaR
- Calculate the actual violation rates (violations/250)
- Compare to the expected rates (5% and 1%)

5.1.2 Theory: The Backtesting Framework

Backtesting compares the VaR a model predicted with the losses that actually followed. A day on which the loss exceeds VaR is a **violation** (also called an exception or a breach).

Key concept: If your α% VaR model is correctly calibrated:

- Expected violations $= n \times (1 - \alpha)$
- Violations should be independent (no clustering)
- The number of violations should follow Binomial$(n, 1 - \alpha)$

Where:

- n = number of days in the test period
- α = confidence level of the VaR

Over 250 days, a 99% VaR should be violated about 2.5 times and a 95% VaR about 12.5 times. No model hits these numbers exactly, so a backtest asks two questions:

1. **Frequency:** Are there too many (or too few) violations to be explained by chance?
2. **Independence:** Are the violations spread evenly, or do they come in clusters?

Basel Traffic Light Zones (250 days, 99% VaR)

Zone	Violations	Multiplier	Interpretation
Green	0-4	3.00	Model acceptable
Yellow	5-9	3.40-3.85	Potential problems, review required
Red	10+	4.00	Model inadequate

The traffic light is a ready-made rule. The regulator has already done the statistics, so the bank only counts its violations and reads off the zone. More violations mean a higher multiplier, and a higher multiplier means more capital.

5.1.3 Activity 2: Basel's Traffic Light System

Task 2.1: Using your violation counts from Activity 1, determine your zone:

- What zone are you in for 99% VaR?
- If your daily VaR is \$10M, what's your capital requirement at each zone? (Simplified: capital = multiplier $\times$ VaR)
- Why do you think Basel uses these specific cutoffs?

5.1.4 Theory: The Binomial Test

Think of each test day as a coin flip with a small chance of landing on "violation": 1% for a 99% VaR. Over 250 days we expect 2.5 violations, but chance alone can give 1, 4 or even 6. The binomial distribution tells us how likely each count is **if the model is correct**.

Step 1: How likely is each count? For 250 days and a 99% VaR:

Violations	Exactly this many	This many or more	Basel zone
0	8.1%	100.0%	Green
1	20.5%	91.9%	Green
2	25.7%	71.4%	Green
3	21.5%	45.7%	Green
4	13.4%	24.2%	Green
5	6.7%	10.8%	Yellow
6	2.7%	4.1%	Yellow
7	1.0%	1.4%	Yellow
8	0.3%	0.4%	Yellow
9	0.1%	0.1%	Yellow
10	0.02%	0.03%	Red

In Excel, for k violations:

- Exactly k: `=BINOM.DIST(k,250,0.01,FALSE)`
- k or more: `=1-BINOM.DIST(k-1,250,0.01,TRUE)`

How to read the table:

- A correct model has 5 or more violations in 10.8% of years, about one year in nine. The yellow zone therefore catches some good models by bad luck.
- A correct model has 10 or more violations in only 0.03% of years. In the red zone, bad luck is no longer a believable excuse.

Step 2: The acceptable range. We reject the model when its violation count is so far from the expected value that a correct model would rarely produce it. `BINOM.INV` gives the range that covers the central 95% of outcomes:

	95% VaR ($p = 0.05$)	99% VaR ($p = 0.01$)
Expected violations	12.5	2.5
Lower bound: `=BINOM.INV(250,p,0.025)`	6	0
Upper bound: `=BINOM.INV(250,p,0.975)`	20	6
Acceptable range at 95% confidence	6 to 20	0 to 6

Note: The 0.025 and 0.975 give us a two-tailed 95% confidence interval (2.5% in each tail).

Step 3: The verdict.

- **Count outside the range:** reject the model at the 5% significance level. A correct model would give such a result less than 5% of the time.
- **Count inside the range:** do not reject the model. This is not proof that the model is right. It only means the data cannot tell it apart from a correct model.

5.1.5 Activity 3: VaR Violations Range

Task 3.1: Answer the following questions

- How likely is your number of 99% violations (or more) if the model is correct?
- Are your violations within the acceptable range?
- If you had 20 violations, what would this suggest about your model?
- If you had only 3 violations, is this good news or bad news?
- What would the acceptable range be if you used 99% confidence for your test?

5.1.6 Theory: The Kupiec Proportion of Failures Test

Understanding the Kupiec Test

Purpose: The Kupiec POF test (1995) asks the same question as the binomial test: is the observed frequency of VaR violations consistent with the model's confidence level? It answers with a single number that is compared with a fixed cutoff.

- **Null Hypothesis (H_0):** The model is correctly calibrated - violation rate equals expected rate (p)
- **Alternative Hypothesis (H_1):** The model is misspecified - violation rate differs from p

The idea: The test compares two explanations of the data:

- the **promised rate** p (1% for a 99% VaR), and
- the **observed rate** $\hat{p} = v/n$.

The observed rate always explains the data at least as well as the promised rate. The test statistic measures how much better. If the model is correct, the two rates are close and the statistic is near zero. The further apart they are, the larger the statistic.

The Test Statistic:

$$LR_{POF} = -2 \ln \left(\frac{(1-p)^{n-v} \times p^v}{(1-\hat{p})^{n-v} \times \hat{p}^v} \right)$$

Where:

- p = expected violation rate (0.01 for 99% VaR, 0.05 for 95% VaR)
- v = observed violations
- n = total observations (250 in our case)
- $\hat{p} = v/n$ (observed violation rate)

In Excel it is easier to calculate the statistic in three parts:

$$\ln L_0 = (n-v)\ln(1-p) + v\ln(p)$$

$$\ln L_1 = (n-v)\ln(1-\hat{p}) + v\ln(\hat{p})$$

$$LR_{POF} = -2\left(\ln L_0 - \ln L_1\right)$$

Where:

- $\ln L_0$ = log-likelihood of the data under the promised rate p
- $\ln L_1$ = log-likelihood of the data under the observed rate $\hat{p}$

If the model is correct, this LR statistic follows a chi-squared distribution with 1 degree of freedom.

Decision Rule:

- If $LR_{POF} > \chi^2_{0.95}(1) = 3.84$, reject the model at 5% significance (In Excel: `=CHISQ.INV(0.95,1)`)
- If $LR_{POF} > \chi^2_{0.99}(1) = 6.63$, reject the model at 1% significance (In Excel: `=CHISQ.INV(0.99,1)`)
- The p-value `=CHISQ.DIST.RT(LR,1)` is the probability of an LR this large or larger when the model is correct. A p-value below 5% leads to the same rejection.

Worked example: A 99% VaR model has $v = 8$ violations in $n = 250$ days.

1. Observed rate: $\hat{p} = 8/250 = 3.2\%$
2. Promised rate: $\ln L_0 = 242 \times \ln(0.99) + 8 \times \ln(0.01) = -39.27$
3. Observed rate: $\ln L_1 = 242 \times \ln(0.968) + 8 \times \ln(0.032) = -35.41$
4. Statistic: $LR_{POF} = -2 \times (-39.27 + 35.41) = 7.73$
5. Verdict: $7.73 > 3.84$, so the model is rejected at 5% significance (p-value 0.5%)

Interpretation:

- The test is **two-sided**: rejects for both too many AND too few violations
- **Small sample problem**: With few expected violations (like 2.5 for 99% VaR), the test has low statistical power - meaning it often fails to detect truly bad models
- The test assumes violations are **independent** - it only checks frequency, not clustering
- With **zero violations** the Excel formula fails, because ln(0) is not defined. Use $LR_{POF} = -2\,n\ln(1-p)$ instead

Note: *More advanced tests like Christoffersen's Conditional Coverage combine frequency and independence testing in one statistic.*

5.1.7 Activity 4: Kupiec's Test Implementation

Task 4.1: Calculate the Kupiec test statistic for your 99% VaR model.

Task 4.2: Test the sensitivity - calculate LR for different violation counts: 0, 2, 5, 7, and 10. What's the minimum violations needed to reject? Maximum allowed?

Task 4.3: Repeat for your 95% VaR model. Which test is more conclusive? Why?

Task 4.4: In Activity 1 you calculated the 99% historical VaR in two ways: with the 3rd worst return and with the percentile function. Both are standard. Compare their backtest results:

- How much do the two VaR figures differ?
- What happens to the violation count, the Basel zone, the binomial test and the Kupiec test?

- Which verdict is the "correct" one?

Task 4.5: Summary:

- Do Basel traffic lights and Kupiec test agree for your model?
- With only 2.5 expected violations, how useful is the Kupiec test for 99% VaR?
- Would you trust this model for setting trading limits?

> **Key Insight**
>
> The Kupiec test often "passes" 99% VaR models even when they're bad - you need 7+ violations (almost 3× expected) to reject! With so few expected violations, a single day can also change the verdict. This is why banks also test at multiple confidence levels.

5.1.8 Activity 5: Independence - Are Violations Clustering?

Task 5.1: List the dates of your violations. Calculate days between consecutive violations. Are they evenly spread or clustered?

Task 5.2: For violation clustering, count:

- How many violations are followed by another violation next day?
- How many violations are followed by no violation?
- If you see several clustered violations, what does this suggest?

Task 5.3: What could cause violation clustering?

> **Why This Matters**
>
> During the 2008 crisis, banks saw weeks of consecutive VaR breaches. Models assuming independent violations drastically underestimated risk during stressed periods.

5.2 Block B: The ES Backtesting Challenge

5.2.1 Theory: Why ES Backtesting is Difficult

> **The ES Backtesting Problem**
>
> Backtesting VaR is a yes-or-no check on every day: was VaR violated or not? ES is the **average loss on the days when VaR is violated**. To test it properly, you need many violation days - but violations are rare by definition.

Key issues:

1. **Few observations:** At 97.5% confidence, expect only ~6 violations per year
2. **Statistical power:** Can't reliably test if the average of 6 observations matches prediction
3. **No simple daily check:** A VaR violation is visible on the day it happens. An average of tail losses has no equally simple daily score (in technical terms, ES is not *elicitable*)

Two practical approaches:

1. **Indirect validation:** Test VaR at multiple confidence levels (90%, 95%, 97.5%)
    - If all levels calibrated → ES likely correct
    - More observations at lower confidence = better statistical power
2. **Direct tests:** Compare realized tail losses to predicted ES
    - Acerbi-Székely test (2014, 2019)
    - McNeil-Frey test (2000)
    - But require strong assumptions and large samples

A simple direct check: Compare the predicted ES with the average loss on the violation days of the test period:

$$\text{Ratio} = \frac{\text{Realized average loss on violation days}}{\text{Predicted ES}}$$

- **Ratio close to 1:** when VaR is violated, losses are about as large as predicted
- **Ratio well above 1:** when VaR is violated, losses are larger than the model said
- With only a few violation days, treat the ratio as a warning sign, not as proof

Regulatory stance: Basel III accepts indirect validation methods because direct ES backtesting remains statistically challenging even with sophisticated tests.

5.2.2 Activity 6: Comprehensive Backtesting - VaR and ES

Use the same split as in Activity 1 (training: days 501-750, testing: days 751-1000) and historical simulation.

Task 6.1: Complete the table:

Level	VaR	Expected	Actual	Predicted ES	Realized Loss	Ratio
95%	?	?	?	?	?	?
97.5%	?	?	?	?	?	?
99%	?	?	?	?	?	?

Expected and Actual are numbers of violations in the test period. Realized Loss is the average loss on the violation days.

> **Excel Hints**
>
> With your VaR in cell `F6`:
>
> - Predicted ES (average of the training losses at or beyond VaR): `=-AVERAGEIF(B502:B751,"<="&-F6)`
> - Realized average loss on violation days: `=-AVERAGEIF(B752:B1001,"<"&-F6)`

Task 6.2: Validation conclusions:

- **VaR validation:** Which levels pass binomial test?
- **ES validation (direct):** Can you trust ES given violation counts?
- **ES validation (indirect):** If all VaR levels calibrated correctly, what about ES?
- **Overall assessment:** Would you use this model for risk management?

5.3 Block C: Stressed Risk Measures

5.3.1 Theory: Stressed Risk Measures

Understanding Stressed Calibration

Context: During the 2007-2008 crisis, banks' VaR models showed low risk because they were calibrated to recent calm periods. When markets crashed, actual losses far exceeded predictions.
Regulatory Response:

- **Basel 2.5 (2011):** Introduced Stressed VaR (SVaR) alongside regular VaR
- **Basel III (2016):** Extended concept to Stressed ES

How it works:

1. **Regular measures:** Calibrated to recent period (typically last 250 days)
2. **Stressed measures:** Calibrated to historical period with highest risk
3. **Capital requirement:** Based on both regular AND stressed measures

Finding the stressed period:

- Test all rolling 250-day windows in your historical data
- Identify window producing highest VaR/ES for current portfolio
- This becomes your "stressed period"

Why it matters:

- Forces banks to hold crisis-level capital even during calm periods
- Prevents procyclicality (risk models becoming optimistic just before crashes)
- Typical stressed measures are 2-4× higher than current measures

Formula evolution:

- Basel II: Capital = $3 \times$ VaR(99%, 10-day)
- Basel 2.5: Capital = $3 \times$ VaR(99%, 10-day) + $3 \times$ SVaR(99%, 10-day)
- Basel III: Capital = $1.5 \times \max$(ES(97.5%, 10-day), Stressed ES(97.5%, 10-day))

5.3.2 Activity 7: Stressed Period and Capital Impact

Download Dataset (2006-2012)

Task 7.1: For each possible 250-day window, calculate 99% VaR and 97.5% ES using historical simulation.

Task 7.2: Identify stressed periods:

Measure	Worst 250-day Window	Value	Recent Window	Value	Ratio
99% VaR	Dates ?-?	?	Dates ?-?	?	?
97.5% ES	Dates ?-?	?	Dates ?-?	?	?

Task 7.3: Calculate capital requirements for $100M portfolio:

Framework	Formula	Recent Period	Stressed Period	Total Capital
Basel II	$3 \times \text{VaR}(99\%)$	?	N/A	?
Basel 2.5	$3 \times \text{VaR} + 3 \times \text{SVaR}$	?	?	?
Basel III	$1.5 \times \max(\text{ES}, \text{SES})$	?	?	?

- How much did capital increase from Basel II to 2.5?
- Why did Basel III reduce the multiplier but still maintain high capital?

> ⚠ Limitation
>
> Your dataset might not contain a true crisis period. Real stressed measures often use 2008 or March 2020 data, showing 3-5× higher risk than calm periods.

5.4 Case Study: Archegos Capital Management (2021)

Background: In March 2021, Archegos Capital, a family office managing $10 billion, collapsed in 48 hours, causing over $10 billion in losses to banks including Credit Suisse ($5.5B), Nomura ($2.85B), and Morgan Stanley ($911M).

The Hidden Leverage Strategy: Archegos used **total return swaps** to secretly build massive positions:

- Banks bought and held the shares, Archegos got profits/losses
- Archegos put up only 10-20% margin, controlled 5-8× more
- No disclosure required (banks owned shares, not Archegos)
- Same positions held at multiple banks - none knew about the others

The Systemic Blind Spot: Each bank thought they had manageable exposure:

- Credit Suisse: "We have $10B exposure, acceptable risk"
- Morgan Stanley: "We have $8B exposure, acceptable risk"
- Nomura: "We have $5B exposure, acceptable risk"

Reality: $30B+ total street exposure to the same 8 stocks.

Why Risk Models Failed:

What VaR Showed	What Actually Happened
Normal volatility in liquid stocks	30% drop when all banks sold together
99% VaR < $50M daily loss	$5.5B loss in 2 days
Adequate 15% margin	Margin calls triggered death spiral
Diversified prime brokerage business	One client destroyed the quarter

The Collapse:

1. ViacomCBS announced share offering → stock dropped 10%
2. Archegos couldn't meet margin calls at ANY bank
3. All banks started selling the same stocks simultaneously
4. Fire sale drove stocks down 30-50%
5. Banks discovered they all had the same positions

Discussion Questions:

1. How would you modify VaR to capture hidden systemic exposure?
2. Should banks be required to disclose large swap positions to each other?
3. If your backtesting shows perfect results but you lose $5B overnight, what assumption was wrong?
4. Could stressed VaR have caught this?

⚠ The Core Lesson

Individual models can be perfect and still fail systemically. Each bank's VaR was "correct" for their known exposure. But swaps created hidden concentration risk across the entire street. When everyone sells the same positions simultaneously, historical correlations and liquidity assumptions collapse.
This wasn't a failure of math - it was a failure to see the whole picture.

5.5 Key Takeaways

- **Backtesting VaR:** Use train-test split, Basel traffic lights (0-4 green, 5-9 yellow, 10+ red), and Kupiec test - but 99% VaR needs 7+ violations to reject (nearly 3× expected)
- **Reading a Backtest:** "Not rejected" does not mean "correct" - with one year of data a single violation can change the verdict, so also check clustering and the size of the losses
- **ES Challenge:** Can't backtest ES directly with few tail events - use indirect validation via multiple VaR levels (90%, 95%, 97.5%) and compare realized tail losses with predicted ES
- **Stressed Measures:** Calculate risk using worst historical 250-day window - typically 2-4× higher than recent period, forcing crisis-level capital in calm times
- **Basel Evolution:** Each crisis increased requirements: Basel II (3×VaR) → Basel 2.5 (3×VaR + 3×SVaR) → Basel III (1.5×max(ES, Stressed ES))
- **Critical Lesson:** Models that pass all backtests can still fail catastrophically (Archegos) - individual models miss systemic risk, and new crisis types won't match historical patterns

6 Test 1: Market Risk Management

6.1 Test Overview

Duration: 70 minutes
Total Points: 25 points

6.2 Test Structure

6.2.1 Part A: Multiple Choice (13 points)

- 26 questions × 0.5 points each
- Four options (a, b, c, d), only one correct
- No negative points for wrong answers
- Covers theoretical concepts from course topics 1-5

6.2.2 Part B: Excel Calculations (12 points)

- 6 calculation problems × 2 points each
- Focus on VaR and ES methods
- Dataset file will be provided: `exam_1.xlsx`
- All calculations must be performed in Excel
- Excel file will be uploaded to the information system showing all calculations with formulas

6.3 Allowed Materials

✅ **Handwritten cheatsheet:** One side of A4 paper, handwritten (not printed), original work only

✅ **Excel:** Use for calculations, must show formulas (no hardcoded values)

✅ **Calculator:** Basic or scientific calculator

❌ **Not allowed:** Printed materials, lecture notes, textbooks, communication devices, internet browsing, AI tools

6.4 Important Notes

- Express all VaR/ES results as positive decimal numbers (e.g., 0.03456)
- Round calculations to 5 decimal places
- Show all work in Excel - formulas must be visible for potential partial credit
- Attempting all questions is better than leaving blanks (no penalties)

Good luck!

7 Reading Week

> **Tip**
>
> No classes this week! Take some rest and use this time to catch up on readings, review course materials, and prepare for the second half of the semester.

8 Liquidity Risk

References

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 21 - Liquidity Risk
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series. - Chapter 6 - Risk Management

Learning Outcomes:

- Distinguish exogenous vs endogenous liquidity risk and quantify liquidation costs from bid-ask spreads and market impact relative to position size
- Evaluate liquidation strategies by comparing patient versus immediate liquidation tradeoffs between liquidity costs and market risk exposure
- Calculate Basel III Liquidity Coverage Ratio (LCR) including HQLA haircuts, stressed cash outflows, and differentiate asset versus funding liquidity risk
- Analyze systemic liquidity crises including margin call mechanics, forced liquidation under leverage, liquidity spirals, and wrong-way risk amplification

8.1 Block A: Understanding Liquidity Risk

You manage a $500M equity portfolio for a mid-sized hedge fund. This morning, you need to liquidate positions in three stocks. The Excel file contains full order books (10 bid levels, 5 ask levels) for three stocks with different liquidity characteristics:

Download Order Book Data

- **TechCorp:** 200,000 share position, 5,000,000 daily volume (highly liquid blue-chip)
- **RegionalBank:** 500,000 share position, 1,200,000 daily volume (moderately liquid mid-cap)
- **BioStartup:** 100,000 share position, 150,000 daily volume (illiquid small-cap)

8.1.1 Activity 1: Discovering the Two Faces of Liquidity

Task 1.1: Small Trade Analysis

Sell **10,000 shares** of each stock at the best available bid price. For each stock, calculate:

- Proceeds from the sale
- Total liquidity cost in $ and % (compared to mid-price)
- Which stock has the highest percentage liquidity cost for this small trade? Why?

Task 1.2: Full Position Liquidation

Now liquidate your **entire position** in each stock by selling at market (hitting successive bid levels in the order book). For each stock, calculate:

- Average execution price
- Total proceeds
- Total liquidity cost in $ and % (compared to mid-price)
- Position size as % of daily volume

Task 1.3: Analysis and Reflection

Compare and analyze your results from Tasks 1.1 and 1.2:

- Rank the three stocks by liquidity cost (%) for both small trades and full liquidation. Did the ranking change? Why?
- For each stock, how did your liquidity cost (%) change when you went from 10,000 shares to your full position? Which stock saw the biggest increase?
- What pattern emerges between position size (as % of daily volume) and total liquidity cost?

Task 1.4: Decomposing Liquidity Cost

Now quantify the two sources of liquidity cost. For each stock, separate your total liquidity cost into:

- **Spread Cost (Exogenous):** The cost if you could execute your entire position at the best bid price
- **Market Impact Cost (Endogenous):** The additional cost from walking down the order book
- Does this decomposition confirm your insights from Task 1.3? How?

8.1.2 Theory: The Two Faces of Liquidity Risk

Let's formalize what you discovered: **exogenous** (market-driven) and **endogenous** (position-driven) liquidity risk.

Liquidity Risk

- **Exogenous Liquidity Risk (Market-Driven):** The liquidity conditions that exist in the market *regardless* of your actions.
  - Bid-ask spread (tightness)
  - Market depth at best quotes
  - Daily trading volume
- **Endogenous Liquidity Risk (Position-Driven):** The liquidity impact *caused by* your own position size.
  - Price impact beyond the spread
  - Your position as % of daily volume
  - Market depth consumption

$$\text{Total Liquidity Cost} = \text{Spread Cost} + \text{Market Impact Cost}$$

- **Spread Cost** = Exogenous (market-driven, constant per share)
- **Market Impact Cost** = Endogenous (position-driven, increasing in position size)

$$\text{Market Impact Cost} = f\left(\frac{\text{Position Size}}{\text{Market Depth}}\right)$$

Risk Management Implications: Never assume you can exit at current market prices—the market is a participant, not a vending machine.

Key Takeaway: Small trades face the market as it is (exogenous), while large trades change the market itself (endogenous). The pattern you observed—cost increasing non-linearly with position size relative to daily volume—is the fundamental reality of liquidity risk.

8.2 Block B: Time Horizons and Liquidation Strategies

Using the same three positions from Block A, you now need to analyze different liquidation strategies. Your risk manager has given you a constraint: **maximum daily participation cannot exceed 20% of normal daily volume** to avoid market disruption.

8.2.1 Activity 2: The Liquidation Tradeoff

Task 2.1: Minimum Liquidation Time

Calculate the minimum number of days needed to liquidate each position under the 20% participation constraint. What does this tell you about position sizing in illiquid markets?

Task 2.2: Patient Liquidation Cost

Assume you use sophisticated execution algorithms that allow the order book to refresh to its normal levels each day. You will liquidate up to 20% of daily volume per day.

- Use the Block A order book to calculate total cost over liquidation period
- How does total liquidity cost compare with immediate liquidation (Block A Task 1.2)?

  Note: This "refresh" assumption is optimistic but helps isolate the benefit of patient liquidation. In reality, the market may react to your selling, which is why execution algorithms and careful timing matter.

Task 2.3: Downside Risk During Liquidation

Risk managers must consider worst-case scenarios. If market volatility is 2% daily for all three stocks, calculate the potential downside risk (1 std dev) from holding positions during gradual liquidation.

For each stock:

$$\text{Downside Risk (1 std dev)} = \text{Position Value} \times \sigma_{\text{daily}} \times \sqrt{\text{Liquidation Days}}$$

where

$$\text{Position Value} = \text{Position Size} \times \text{Mid Price (from Block A)}$$

This represents the potential loss if the market moves **against you** during the liquidation period (note: markets can also move in your favor, but prudent risk management focuses on downside protection).

Calculate:

- Downside risk exposure for each stock
- Liquidity cost savings from patient liquidation: (Block A Task 1.2 cost) - (Block B Task 2.2 cost)
- Risk-adjusted benefit: Cost Savings - Downside Risk (if positive, patient liquidation remains beneficial even in the worst-case scenario)

Task 2.4: Analysis - The Liquidation Tradeoff

Compare your results from Tasks 2.1, 2.2, and 2.3:

- Which stock has the largest liquidity cost savings from patient liquidation (in absolute $ and % terms)?
- For which stock(s) is the risk-adjusted benefit positive? For which is it negative? What does this tell you about when patient liquidation makes sense?
- If you were the portfolio manager facing potential adverse market movements, which liquidation strategy (immediate vs patient) would you choose for each stock and why?
- How do your Block A findings (exogenous vs endogenous costs from Task 1.4) explain why patient liquidation's risk-adjusted benefit varies across stocks? (Hint: Which type of cost is reduced by patient liquidation?)

8.2.2 Theory: The Liquidation Timeline Framework

- **Immediate Liquidation:** Eliminates market risk quickly — but incurs maximum liquidity costs (spread + severe market impact).
- **Gradual Liquidation:** Reduces market impact — but increases exposure to adverse price movements over the liquidation horizon.

$$\text{Minimum Liquidation Time} = \frac{\text{Position Size}}{\text{Daily Volume} \times \text{Max Participation Rate}}$$

Practical Guidelines

Participation Rate	Market Signal	Typical Impact
$< 5\%$	Invisible	Minimal
5-10%	Normal trading	Slight
10-20%	Large player	Noticeable
20-30%	Market mover	Significant
$> 30\%$	Distressed seller	Severe

During market stress:

- Daily volumes can drop 50-70%
- Bid-ask spreads widen 3-10x
- Market depth evaporates
- **Result:** A 2-day liquidation becomes a 2-week ordeal

Key Insight: Liquidity risk is non-linear and procyclical - it's minimal when you don't need it and severe when you do.

8.2.2.1 Liquidity-Adjusted VaR

Traditional VaR assumes positions can be liquidated instantaneously at market prices. Liquidity-adjusted VaR (LVaR) incorporates the cost of unwinding positions over a realistic time horizon:

$$\text{LVaR} = \text{VaR} + \text{Liquidation Cost}$$

Banks use internal models to estimate liquidation costs based on historical bid-ask spreads, market depth, and trading volumes. These models typically:

1. **Assess position size relative to market liquidity** - larger positions relative to daily volume face higher market impact
2. **Determine realistic liquidation horizon** - the time needed to unwind without excessive market disruption (often capped at 10-20% daily participation rate)
3. **Estimate total execution cost** - combining immediate spread costs with projected market impact over the liquidation period

The liquidation cost component in LVaR reflects the expected cost under the chosen liquidation strategy (immediate vs gradual), calibrated to historical market conditions and stressed scenarios.

Key insight: A portfolio may have acceptable VaR but unacceptable LVaR if positions are large relative to market liquidity. Liquidity risk dominates market risk for illiquid positions.

8.3 Block C: Funding Liquidity Risk and the Liquidity Coverage Ratio (LCR)

So far, we've focused on **asset liquidity** - can you sell assets without major price impact? Now we turn to **funding liquidity** - can you meet cash obligations as they come due?

Banks face a unique challenge: liabilities (deposits) can flee rapidly during stress, but assets (loans) cannot be recalled quickly. If depositors demand $10B in withdrawals today, you cannot say "wait 30 years for these mortgages to mature." You need liquid assets - a buffer - to meet sudden outflows.

The 2008 financial crisis revealed that many banks held insufficient liquid assets relative to their unstable funding. Within days, wholesale funding markets froze, and banks faced severe funding runs. In response, Basel III introduced the **Liquidity Coverage Ratio (LCR)** to ensure banks maintain adequate high-quality liquid assets to survive a 30-day acute stress scenario.

> **Key Concepts for This Activity**
>
> **High-Quality Liquid Assets (HQLA):** Assets that can be quickly converted to cash with minimal loss of value, even during market stress. Basel III defines three levels:
>
> - **Level 1 (0% haircut):** Cash, central bank reserves, government bonds - the most liquid, can be sold at full value even in crisis
> - **Level 2A (15% haircut):** High-quality corporate bonds, covered bonds - liquid but

might sell at 85 cents on the dollar in stress
- **Level 2B (25-50% haircut):** Lower-rated corporates, equities - less liquid with higher haircuts in stress

Haircuts: Percentage reduction in market value to reflect stressed liquidation conditions. A 15% haircut means a $100 bond counts as only $85 of available liquidity.
Liquidity Coverage Ratio (LCR): Basel III requirement that banks hold sufficient HQLA to cover 30 days of stressed net cash outflows:

$$\text{LCR} = \frac{\text{HQLA}}{\text{Net Cash Outflows over 30 days}} \times 100\% \geq 100\%$$

Stressed Cash Outflows: Basel III assumes liability outflow rates calibrated to the 2008 crisis - stable retail deposits are sticky (5% outflow), but wholesale funding runs immediately (100% outflow).

8.3.1 Activity 3: Computing the Liquidity Coverage Ratio

You're the treasurer of MidSize Bank. Your simplified balance sheet shows:

Assets	Liabilities
Cash & Central Bank Reserves: $5B	Stable Retail Deposits: $60B
Government Bonds (AAA): $15B	Less Stable Retail Deposits: $20B
Corporate Bonds (Investment Grade): $20B	Corporate Deposits (operational): $15B
Corporate Loans: $50B	Corporate Deposits (non-operational): $10B
Retail Mortgages: $40B	Wholesale Funding (< 30 days): $25B

Task 3.1: Calculate High-Quality Liquid Assets (HQLA)

Calculate the total HQLA by applying the appropriate haircuts to the balance sheet assets.

Task 3.2: Calculate 30-Day Stressed Cash Outflows

Calculate expected cash outflows over 30 days using these outflow rates:

- Stable Retail Deposits: 5%
- Less Stable Retail: 10%
- Corporate Operational: 25%
- Corporate Non-operational: 40%
- Wholesale Funding: 100%

Task 3.3: Calculate the LCR

Calculate MidSize Bank's LCR and determine:

- Does it meet the regulatory minimum of 100%?
- What is the buffer (or shortfall)?

Task 3.4: Crisis Stress Test

Now assume a severe market crisis occurs:

- Government bonds lose 10% of market value
- Corporate bond haircuts increase from 15% to 50%

Recalculate:

- New HQLA value
- New LCR
- If LCR $< 100\%$, what is the funding gap in dollars?

Task 3.5: Funding Liquidity vs Asset Liquidity

Compare your Block C findings with Blocks A and B:

- How does funding liquidity risk (Block C) differ from asset liquidity risk (Blocks A & B)?
- In Task 3.4, which asset class deterioration caused the biggest HQLA decline? Why does this matter for bank risk management?
- MidSize Bank has \$25B in wholesale funding (19% of liabilities). How is reliance on wholesale funding similar to holding large illiquid positions (like BioStartup in Block A)?
- Can a bank have good asset liquidity but still fail from funding liquidity crisis? Explain using your findings.

8.3.2 Theory: Funding Liquidity Risk and Basel III LCR

8.3.2.1 Funding Liquidity vs Asset Liquidity

Asset Liquidity (Blocks A & B)

- Can I sell my assets without major price impact?
- Focus: Bid-ask spreads, market depth, position size

Funding Liquidity (Block C)

- Can I meet my cash obligations as they come due?
- Focus: Liability stability, rollover risk

The Connection: Funding stress forces asset fire sales (selling at Block A prices with high market impact), which depletes HQLA, which triggers more funding stress.

8.3.2.2 Liquidity Metrics

$$\text{LCR} = \frac{\text{HQLA}}{\text{Net Cash Outflows over 30 days}} \times 100\% \geq 100\%$$

The LCR ensures banks can survive acute stress for 30 days. Basel III also requires the **Net Stable Funding Ratio (NSFR)** for long-term structural stability:

$$\text{NSFR} = \frac{\text{Available Stable Funding}}{\text{Required Stable Funding}} \geq 100\%$$

Two dimensions of funding liquidity:

- **LCR:** Can you survive a 30-day crisis? (Short-term shock absorption)
- **NSFR:** Is your funding stable for your asset maturity? (Long-term structural stability)

A bank might pass LCR but fail NSFR if it funds long-term assets (e.g., 30-year mortgages) with short-term liabilities (e.g., overnight wholesale funding). LCR helps survive the crisis; NSFR helps avoid the crisis.

8.3.2.3 The Death Spiral

1. **Rumors of problems** → Wholesale funding dries up
2. **Sell liquid assets** → HQLA depletes
3. **Fire sale illiquid assets** → Losses mount
4. **More rumors** → Retail deposit run
5. **Bank failure**

Key Insight: Banks don't fail from bad loans (that takes years). They fail from funding runs (that takes days). Your Task 3.4 stress test showed how quickly this can happen—HQLA evaporates when you need it most.

8.4 Block D: Case Study - The GameStop Short Squeeze (January 2021)

In January 2021, GameStop (GME) became the center of an unprecedented liquidity crisis that combined all the elements you've studied: asset liquidity evaporation, funding liquidity stress, forced liquidation, and systemic feedback loops.

The Setup: GameStop, a struggling video game retailer, had become heavily shorted by hedge funds—short interest exceeded 100% of the available shares. Retail traders on Reddit's WallStreetBets forum coordinated buying, triggering a massive short squeeze. Between January 13-28, the stock surged from $20 to $483.

What happened to liquidity? As the price soared, multiple liquidity crises unfolded simultaneously:

- **Asset liquidity vanished:** Bid-ask spreads that were normally $0.01-0.05 widened to $10-50 during peak volatility. Market makers stepped back, and trading volume that was normally 7 million shares/day spiked to over 175 million shares/day—yet liquidity actually decreased as spreads exploded.
- **Funding liquidity crisis:** Robinhood, the popular trading app used by retail traders, faced a liquidity crisis when the Depository Trust & Clearing Corporation (DTCC) increased margin requirements from $700 million to $3 billion overnight. Unable to meet this demand immediately, Robinhood restricted trading, triggering outrage and congressional hearings.
- **Hedge fund forced liquidation:** Short sellers like Melvin Capital faced catastrophic losses and margin calls. To close their short positions, they had to **buy** shares—but each purchase pushed the price higher, triggering more margin calls. They couldn't use patient liquidation because **margin calls demanded immediate action**.

8.4.1 Activity 4: Analyzing the GameStop Liquidity Crisis

Reflect on the GameStop case and connect it to your learnings from Blocks A, B, and C:

Task 4.1: Asset Liquidity Breakdown (Block A Connection)

- During normal times, GME had tight spreads and reasonable depth. During the squeeze, spreads widened 100-1000x and depth evaporated. Which type of liquidity risk dominated initially (exogenous or endogenous)? How did it change as hedge funds were forced to cover shorts?

Task 4.2: The Impossible Liquidation Timeline (Block B Connection)

- In Block B, you learned that patient liquidation reduces costs but exposes you to market risk. Why couldn't short sellers use patient liquidation for GME? What happens to the Block B tradeoff when you face margin calls?
- The squeeze lasted about 2 weeks. If short sellers had time to liquidate over 30 days at 20% of daily volume (Block B constraint), would this have prevented the crisis? Why or why not?

Task 4.3: Funding Liquidity Crisis (Block C Connection)

- Robinhood faced a "margin call" when DTCC increased requirements from $700M to $3B. How is this similar to a bank failing to meet its LCR requirement (Block C)? What was Robinhood's equivalent of HQLA?
- DTCC's margin requirements are procyclical—they increased precisely when volatility spiked. How does this compare to the haircut increases in Block C Task 3.4? What does this reveal about liquidity stress in the financial system?

Task 4.4: Liquidity Wrong-Way Risk and Black Holes

- Short sellers needed to BUY to close positions, but each purchase pushed prices higher, triggering more margin calls and more forced buying. Trace this feedback loop and explain why this is the perfect example of liquidity wrong-way risk.
- At the peak, GME became a "liquidity black hole"—normal price discovery broke down, spreads were enormous, and market makers couldn't provide continuous pricing. What could have broken this cycle? What actually did break it?

8.4.2 Theory: Systemic Liquidity Concepts

8.4.2.1 Liquidity Wrong-Way Risk

Wrong-way risk occurs when liquidity needs increase precisely when asset liquidity deteriorates. The GameStop short squeeze is a textbook example:

- **Funding needs spiked:** Short sellers faced massive margin calls requiring immediate cash
- **Asset liquidity vanished:** Bid-ask spreads exploded, market makers stepped back
- **Timing correlation:** Those who needed to buy the most faced the worst execution costs
- **Feedback loop:** Each forced purchase made the problem worse for remaining shorts

This adverse correlation amplifies losses beyond what separate funding and market risk models would predict.

8.4.2.2 Liquidity Black Holes

A liquidity black hole is an extreme state where market liquidity completely evaporates:

- **No continuous pricing:** During GME's peak, normal bid-ask quotes disappeared for seconds or minutes at a time
- **Self-reinforcing dynamics:** Each forced purchase triggered more margin calls, creating more forced purchases

- **Market freeze:** Normal price discovery broke down—prices jumped in $50-100 increments rather than pennies

Historical examples: October 1987 crash, September 2008 (Lehman), March 2020 (COVID panic), **January 2021 (GameStop)**. Once triggered, standard risk measures become unreliable because they assume continuous markets and normal liquidity.

8.4.2.3 The GameStop Lesson

The GameStop crisis revealed several crucial insights:

1. **Position limits matter:** Short interest >100% of float created structural vulnerability
2. **Procyclical margin requirements:** DTCC's margin increases amplified the crisis
3. **Concentration risk:** Retail flow concentrated through few platforms (Robinhood) created single points of failure
4. **Social media coordination:** New dynamics in market manipulation and coordination
5. **Liquidity is not continuous:** Even major stocks can become illiquid under stress

8.5 Key Takeaways

This module revealed how liquidity crises unfold through three interconnected dimensions, culminating in the GameStop short squeeze—a real-world laboratory for liquidity risk.

The Three Dimensions of Liquidity:

- **Asset Liquidity (Blocks A & B):** Markets can become illiquid from external conditions (exogenous risk) or from your own trades (endogenous risk). Time is a luxury—patient liquidation only works when you face no margin calls.
- **Funding Liquidity (Block C):** Banks survive on stable funding, not just assets. LCR measures whether you can survive 30 days of stress. The death spiral: rumors → funding dries up → forced asset sales → more rumors → failure.
- **Systemic Liquidity (Block D):** The GameStop case showed how all three dimensions interact catastrophically. Short sellers faced wrong-way risk (needed to buy when liquidity vanished), procyclical margins (DTCC requirements tripled overnight), and a liquidity black hole (spreads widened 1000x, price discovery failed).

The Bottom Line: Liquidity risk is not about having assets—it's about converting them to cash when you need it most, which is precisely when markets least want to provide it.

9 Credit Risk

References

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 17 - Estimating Default Probabilities
  - Chapter 18 - xVAs
  - Chapter 19 - Credit Value at Risk

Learning Outcomes:

- Calculate expected loss (PD × LGD × EAD) and distinguish between expected and unexpected loss for capital management
- Analyze credit migration risk: calculate mark-to-market losses from rating downgrades and identify fallen angel effects
- Measure default probabilities using cumulative, marginal, and hazard rate approaches, and interpolate to any maturity
- Extract market-implied default probabilities from CDS spreads and explain why they exceed historical rates
- Connect credit risk to liquidity risk through fallen angel HQLA disqualification and wrong-way risk

9.1 Block A: Understanding Credit Risk Components

9.1.1 Theory: The Three-Component Framework

Credit risk differs fundamentally from market risk. While market risk involves continuous, observable price movements that are relatively symmetric, credit risk is **asymmetric** (limited upside vs. potential total loss), **discrete** (default or no default), and **latent** (deterioration invisible until default).

To quantify credit risk, we decompose every exposure into three multiplicative components:

- **Probability of Default (PD):** Likelihood of borrower failure within one year (estimated from historical rates, credit ratings, or internal models).
- **Loss Given Default (LGD):** Percentage lost if default occurs (depends on collateral quality and seniority).

$$\text{LGD} = 1 - \text{Recovery Rate}$$

- **Exposure at Default (EAD):** Dollar amount at risk when default occurs. For credit lines:

$$\text{EAD} = \text{Current Exposure} + (\text{Undrawn} \times \text{CCF})$$

  where CCF (Credit Conversion Factor) captures borrower drawdown behavior before default.

These combine multiplicatively to produce **Expected Loss**:

$$\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD}$$

Expected loss (EL) is the **mean** of the loss distribution—predictable and priced into loan spreads. **Unexpected loss** (UL) is the **volatility** around that mean—the true risk requiring capital protection. This separation drives banking economics: spreads must cover EL, while capital absorbs UL.

Connection to Week 8: Credit downgrades create wrong-way risk for liquidity—collateral backing funding facilities loses value precisely when funding stress peaks.

9.1.2 Activity 1: Loan Portfolio Analysis

You are a loan officer at Regional Bank evaluating a portfolio of 10 loans. Calculate expected loss for each loan using the three-component framework.

Download Credit Risk Data

Use the **Loan_Portfolio** sheet from the Excel file.

Task 1.1: Calculate PD, LGD, and EAD

For each of the 10 loans, calculate:

- **PD:** Use the Historical_Default_Rate_Pct column
- **LGD:** Calculate as $\max(0, \frac{\text{Exposure}-\text{Collateral Value}}{\text{Exposure}})$ (assumes collateral recovers at stated value; in practice, banks apply haircuts and workout costs)
- **EAD:** Apply the formula above with CCF = 50% (the Basel III standard for committed credit lines)

Then analyze patterns:

- Which borrower type has the highest average PD? Why might BB-rated Corporate (L002) and BB-rated SME (L006) have different PDs despite identical ratings?
- Compare LGD for residential mortgages L007 vs. L009. What drives the difference? What happens when collateral exceeds exposure?
- Which loans have EAD > Current Exposure, and why does this matter for risk management?

Task 1.2: Calculate Expected Loss

Combine the three components: EL = PD × LGD × EAD. Calculate both dollar EL and EL as % of EAD.

- Which loan has the highest dollar EL? Highest EL percentage? What drives these differences?
- Compare L001 (BBB Corporate, Equipment) vs. L010 (AAA Corporate, Government bonds). Which has higher EL and why?
- If you could only monitor 3 loans intensively, which would you choose based on EL and component analysis?

9.1.3 Key Insights: EL vs. UL and Banking Economics

Expected Loss (EL): The predictable mean loss that should be covered by loan pricing (interest rate spreads) and provisions. If mispriced, banks lose money systematically even without defaults.

Unexpected Loss (UL): The volatility around EL (standard deviation) representing true credit risk. Covered by bank capital (equity). For regulatory purposes:

$$\text{Credit VaR}_{99.9\%} = \text{EL} + z_{99.9\%} \times \text{UL}$$

Recovery rates vary significantly by collateral type. Industry benchmarks: government bonds (95%), residential property (70%), equipment (50%), unsecured (10%). Banks apply haircuts to collateral values and add workout costs when estimating regulatory LGD.

Risk management implications: The multiplicative formula allows component substitution—improve any component to reduce EL. Different mitigation strategies target different components: stricter underwriting (lower PD), collateral requirements (lower LGD), exposure limits (lower EAD).

This framework supports Basel III capital requirements (covered in later blocks): regulatory capital must cover unexpected losses at 99.9% confidence, while provisions cover expected losses. The separation between pricing (EL) and capital (UL) is fundamental to banking profitability.

9.2 Block B: Credit Migration and Mark-to-Market

9.2.1 Theory: Credit Risk Beyond Default

Block A focused on default risk—borrowers failing to repay. But **credit risk is not binary**. You can lose money even without defaults occurring, whenever credit quality deteriorates.

Three Sources of Credit Losses:

- **Default Risk:** Borrower fails to repay (binary event, covered in Block A)
- **Migration Risk:** Credit rating downgrades reduce bond values through spread widening
- **Spread Risk:** Market-wide credit spread movements independent of specific rating changes

Why Migration Matters: For **trading books** (mark-to-market accounting), migration and spread risk dominate daily P&L. For **banking books** (held-to-maturity accounting), only default matters if bonds are held to maturity without impairment.

The Bond Pricing Mechanism: When credit quality deteriorates and rating agencies downgrade bonds, market participants demand higher yields to compensate for increased risk:

$$\text{Bond Yield} = \text{Risk-Free Rate} + \text{Credit Spread}$$

As spreads widen, bond prices must fall to bring yields up to market levels:

$$\Delta\text{Price}\% \approx -\text{Modified Duration} \times \Delta\text{Spread}$$

The Fallen Angel Effect: When bonds cross from investment grade (BBB- or better) to high yield (BB+ or worse), they trigger institutional forced selling (pension funds, insurance

mandates cannot hold high-yield bonds), HQLA disqualification for LCR (Week 8 connection), and liquidity evaporation—creating losses beyond normal spread widening.

9.2.2 Activity 2: Bond Portfolio Revaluation

You are a fixed income portfolio manager at Investment Fund. This quarter, rating agencies announced downgrades for several bonds in your portfolio. Assess the mark-to-market impact.

Use the **Bond_Portfolio** sheet from the data file.

Task 2.1: Calculate Mark-to-Market Losses

For each bond with a rating change:

- Calculate the new bond yield using the updated credit spread (from Credit_Spreads table based on new rating and maturity)
- Price the bond using: `=-PV(new_yield, maturity, coupon, 100)` in Excel (assumes annual coupons)
- Calculate the mark-to-market loss in dollars

Then analyze:

- Which bond experiences the largest dollar loss from rating migration?
- Compare percentage losses: are longer-maturity or shorter-maturity bonds more affected by identical rating downgrades? Why does this pattern emerge?

Task 2.2: Fallen Angels and Liquidity Impact

- Which bond(s) crossed from investment grade (BBB- or better) to high yield (BB+ or worse)?
- Why is this transition especially problematic? What institutional consequences does it trigger?
- If you hold the fallen angel to maturity without default, will you recover the MTM loss?
- Which accounting regime makes rating downgrades more painful: trading books or banking books?
- Before the downgrade, the fallen angel qualified for your LCR at 85% value. What happens after it becomes high yield?
- Explain the wrong-way risk: why do credit downgrades create maximum problems during stress periods?

9.2.3 Key Insights: Migration Risk and Portfolio Management

Trading vs. Banking Books: Trading books mark-to-market daily, so rating migrations create immediate P&L losses and capital charges. Banking books use amortized cost—if held to maturity without default, MTM losses don't materialize. However, banks must recognize impairment if credit deterioration becomes permanent under IFRS 9/CECL.

Duration Amplifies Credit Risk: Longer-duration bonds suffer larger losses from identical spread widening. A 10-year bond (duration 7.5) loses ~7.5% when spreads widen 100 bps, while a 2-year bond (duration 1.9) loses only ~1.9%. This explains why duration management is crucial for credit portfolios.

Wrong-Way Risk (Week 8 Callback): Credit downgrades trigger wrong-way risk—collateral backing funding facilities loses value precisely when liquidity is needed most. Fallen angels lose

HQLA eligibility entirely, forcing fire sales during stress periods and creating liquidity doom loops.

9.3 Block C: Historical Default Probabilities

9.3.1 Theory: Measuring Default Probability

When analyzing credit risk, we use three interconnected measures that provide equivalent information from different perspectives:

1. Cumulative (Unconditional) Default Probability - $Q(t)$

- Probability of default within $[0, t]$.
- "What's the chance they default sometime in the next t years?"
- Always increases with time.
- Example: $Q(5) = 1.54\%$ means 1.54% of BBB companies default within 5 years.

2. Marginal (Conditional) Default Probability - $q(t)$

- Probability of default in period t, given survival through $t-1$.
- "If they survived this long, what's the chance they default next period?"
- Can increase or decrease over time.
- Formula:

$$q(t) = \frac{Q(t) - Q(t-1)}{1 - Q(t-1)}$$

3. Hazard Rate - $h(t)$

- Instantaneous default intensity (continuous-time version of marginal PD).
- Used to interpolate default probabilities to any maturity.
- For constant hazard rate:

$$Q(t) = 1 - e^{-h \cdot t}$$

$$h = -\frac{1}{t} \ln(1 - Q(t))$$

Additionally, **Credit Rating Transition Matrices** show one-year probabilities of rating changes, capturing both defaults (in the default column) and migrations (downgrades/upgrades) simultaneously.

9.3.2 Activity 3: Historical Default Probability Analysis

Use the **Default_Probs** sheet from the data file. It contains actual default data from rating agencies (1981-2020).

Task 3.1: Interpreting and Calculating Default Probabilities

- What percentage of BBB companies default within 5 years? How many defaults would you expect from 100 B-rated bonds held for 2 years?
- Calculate marginal default probabilities for BBB-rated companies for years 2 and 5.
- Compare marginal PDs for year 5: Calculate for both AA and B ratings. What pattern emerges as credit quality deteriorates?

Task 3.2: Hazard Rates and Non-Integer Maturities

Banks need default probabilities for any time horizon, not just whole years.

- Calculate the constant hazard rate for AA companies using the 5-year cumulative PD.
- Using this hazard rate, calculate cumulative default probabilities for 2.5-year and 7-year maturities.
- Repeat for BB-rated companies. How do the hazard rates compare across credit quality?

9.3.3 Key Insights: Default Probability Measures and Transition Matrices

The Three Measures Are Mathematically Equivalent: Cumulative PD, marginal PD, and hazard rates contain the same information viewed differently. Banks choose based on use case: cumulative PD for total exposure risk, marginal PD for time-varying risk patterns, hazard rates for pricing and interpolation.

Hazard Rates Enable Flexibility: The constant hazard rate assumption allows banks to extract default probabilities for ANY maturity from limited historical data. While real hazard rates aren't perfectly constant, this approximation works well for most credit applications.

Rating-Specific Patterns: Investment-grade ratings (AAA-BBB) show low, gradually increasing marginal PDs. High-yield ratings (BB-B) show high initial marginal PDs that may decline over time (survivors become stronger). CCC/C ratings have extreme first-year PDs reflecting near-term distress.

Credit Rating Transition Matrix Applications: The transition matrix provides one-year migration probabilities AND one-year default probabilities simultaneously. Diagonal values show rating stability. Banks use this for: portfolio loss estimation from defaults AND downgrades, regulatory stress testing, fallen angel probability (BBB $\rightarrow$ BB transition), and MTM loss calculations.

9.4 Block D: Market-Implied Default Probabilities

9.4.1 Theory: From CDS Spreads to Default Probabilities

Block C showed historical default probabilities from past data (1981-2020). But markets are forward-looking: **CDS spreads** tell us what investors expect going forward and what they charge for default protection today.

Credit Default Swaps (CDS): Insurance contracts against default. The buyer pays an annual premium (the CDS spread, in basis points) to transfer default risk to the seller. Higher expected default probability $\rightarrow$ higher CDS spread.

Risk-Neutral vs. Historical Probabilities:

- **Historical (Real-World) PD:** Actual default frequencies from rating agency data (Block C)
- **Risk-Neutral (Market-Implied) PD:** Default probabilities extracted from CDS market prices
- **Why they differ:** Risk-neutral PDs systematically exceed historical PDs because markets price in risk premiums—investors demand extra compensation beyond expected loss for bearing default risk

Extracting Default Probabilities from CDS Spreads:

Using Block C's constant hazard rate framework applied to market prices: Given CDS spread s (as decimal) and recovery rate R, the implied hazard rate is:

$$h = \frac{s}{1-R}$$

The risk-neutral cumulative default probability over t years is:

$$Q^{RN}(t) = 1 - e^{-h \cdot t} = 1 - e^{-\frac{s \cdot t}{1-R}}$$

Example: Ford CDS spread = 180 bps = 0.0180, recovery = 40%, so

$$h = 0.0180/0.60 = 0.03$$

$$Q^{RN}(5) = 1 - e^{-0.03 \times 5} = 13.9\%$$

9.4.2 Activity 4: Extracting Market-Implied Probabilities

Current 5-Year CDS Spreads (basis points):

Company	Rating	CDS Spread
Apple	AA	35 bps
IBM	A	65 bps
Ford	BBB	180 bps
Tesla	BB	420 bps
AMC	B	850 bps

Assume 40% recovery rate for all companies.

Task 4.1: Calculate and Compare Market-Implied vs. Historical Probabilities

For IBM, Ford, and Tesla:

- Calculate the implied hazard rate.
- Calculate the 5-year risk-neutral default probability.
- Compare to historical 5-year PDs from Block C (A: 0.46%, BBB: 1.54%, BB: 6.43%)
- What pattern emerges? Why do market-implied PDs consistently exceed historical rates?
- Calculate the risk premium (risk-neutral PD − historical PD) for each company. Which rating shows the largest premium in percentage points?

Task 4.2: Bond-CDS Arbitrage Analysis

Ford's 5-year bond yields 5.2% while the 5-year Treasury yields 3.5%, giving a bond credit spread of 170 bps. Ford's CDS spread is 180 bps.

- In theory, bond spreads and CDS spreads should converge (both price default risk). Why might they differ by 10 bps in practice?
- If Ford's CDS spread jumped to 250 bps while the bond spread stayed at 170 bps, what arbitrage trade could you execute?

9.4.3 Key Insights: Market-Implied Probabilities and the Risk Premium

Risk-Neutral > Historical: Market-implied default probabilities systematically exceed historical default rates because CDS prices embed risk premiums. Investors demand compensation beyond expected loss for bearing default risk—especially for lower-rated credits where losses are concentrated and correlation risk is high. The risk premium reflects investor risk aversion and market liquidity conditions.

Forward-Looking vs. Backward-Looking: Historical PDs tell us what happened over 1981-2020 across all economic cycles. CDS spreads reflect current market expectations given today's economic conditions, credit cycle position, and investor risk appetite. During stress periods (2008, 2020), CDS spreads spike far above long-run historical averages.

Bond-CDS Basis: In frictionless markets, bond credit spreads and CDS spreads should align (both price default risk). Deviations create arbitrage opportunities through basis trades. Persistent basis reflects funding costs, counterparty risk in CDS contracts, and cheapest-to-deliver options in bond delivery.

Practical Applications: Banks use market-implied PDs for trading book valuation (mark-to-market), credit portfolio hedging, and counterparty credit risk (CVA calculations). Historical PDs are used for banking book provisions (IFRS 9/CECL) and stress testing. Connection to Block B: CDS spreads and bond spreads move together, both reflecting migration risk and default risk simultaneously.

9.5 Key Takeaways

- **Expected loss** (PD × LGD × EAD) measures average credit losses, while **Credit VaR** captures unexpected losses that determine capital requirements
- **Recovery rates** vary significantly by seniority and collateral, with senior secured bonds recovering substantially more than subordinated debt
- **Credit migration risk** creates mark-to-market losses even without default, as rating downgrades widen spreads and reduce bond values
- **Default probabilities** can be measured three ways (cumulative, marginal, hazard rate) and allow calculation of default risk for any time horizon
- **Historical default rates** increase sharply with lower credit ratings, demonstrating the nonlinear relationship between rating quality and default risk
- **Market-implied probabilities** from CDS spreads exceed historical default rates because they include risk premiums and reflect forward-looking investor expectations

10 Operational Risk

References

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 20 - Operational Risk

Learning Outcomes (Master Level):

- Categorize operational losses using Basel's framework and analyze their frequency–severity patterns
- Calculate Standardized Measurement Approach (SMA) capital requirements, including sensitivity to loss experience
- Design and critically assess Key Risk Indicators (KRIs) within an RCSA framework

10.1 Block A: Understanding Operational Loss Patterns

10.1.1 Theory: Operational Risk Definition

Operational risk refers to potential losses from failed internal processes, human errors, system failures, or external events. Unlike credit or market risk, operational risk is challenging to quantify due to its diverse and unpredictable nature.

Basel Definition

"The risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events."

- **Includes**: Legal risk (lawsuits, regulatory penalties)
- **Excludes**: Reputation risk, strategic decision risks

10.1.1.1 The Seven Basel Categories

Basel III classified operational risks into seven distinct types:

1. **Internal Fraud**: Intentional acts by employees to deceive or circumvent laws. *(Examples: Unauthorized trading, insider trading, embezzlement)*
2. **External Fraud**: Fraudulent activities by third parties *(Examples: Cyber attacks, phishing, robbery)*
3. **Employment Practices and Workplace Safety**: Violations of employment laws *(Examples: Discrimination lawsuits, wrongful termination, workplace injuries)*

4. **Clients, Products, and Business Practices**: Failures in professional obligations or product design *(Examples: Regulatory violations, data breaches, money laundering, mis-selling)*
5. **Damage to Physical Assets**: Loss or damage from natural disasters or external events *(Examples: Fires, floods, earthquakes, terrorism)*
6. **Business Disruption and System Failures**: Disruption from IT or infrastructure failures *(Examples: System outages, cyberattacks, hardware failures, power outages)*
7. **Execution, Delivery, and Process Management**: Failed transaction processing or process management *(Examples: Data entry errors, settlement failures, payment processing errors)*

10.1.2 Activity 1: Loss Data Analysis

You are a newly appointed operational risk analyst at a major European bank. Your risk committee has asked you to analyze the bank's operational loss history over the past decade.

Download Operational Risk Data

Open the Excel file and navigate to **Activity_1_Loss_Data** sheet.

Task 1.1 Focus first on the **top 7 loss events**, where the Basel category column is intentionally left blank. Categorize each of these 7 operational losses using Basel's seven categories.

Note: Some losses may seem to fit multiple categories—use your judgment to identify the primary operational risk cause.

Task 1.2 After completing Task 1.1 (i.e., filling in the Basel categories for the top 7 rows), use the **full dataset** (all rows in the sheet) to quantify and interpret patterns across Basel categories:

- In Excel, compute for each Basel category: (i) number of events, (ii) average loss severity, and (iii) share of total losses.

Then discuss:

- Which categories show **high-frequency, low-severity** patterns?
- Which categories show **low-frequency, high-severity** patterns?
- What implications do these patterns have for risk management strategy? Which category would you prioritize for controls vs. capital?

Task 1.3 Looking at the distribution of all loss amounts in the full dataset:

- How would you describe its shape (symmetric vs. skewed, thin-tailed vs. heavy-tailed)?
- Compute the mean and standard deviation and compare them to the median and 95th percentile – what do these summary statistics tell you about tail risk?
- Why does this matter for operational risk modeling and for capital requirements?

10.1.3 Theory: Loss Frequency and Severity Distributions

Your analysis in Activity 1 revealed a fundamental characteristic of operational risk: losses are **not normally distributed**. Most losses are small, but a few are catastrophic—this is known as a **heavy-tailed distribution**.

From a modeling perspective, operational losses are often treated using a **loss distribution approach (LDA)**: separate models for how often losses occur and how large they are when they occur, combined into an annual loss distribution. Even without running a full LDA model, it is useful to explicitly distinguish between two dimensions:

- **Frequency**: How often do losses occur? (daily data entry errors vs. rare cyber breaches)
- **Severity**: How large are losses when they occur? (€500K settlement vs. €85M hacking incident)

This creates four operational risk profiles that you can already see in your Excel summary from Activity 1:

Profile	Frequency	Severity	Management Approach	Examples
High-Frequency, Low-Severity	Common	Small	Process improvement, automation	System outages, processing errors
Low-Frequency, Low-Severity	Rare	Small	Accept or transfer (insurance)	Phishing attacks, minor fraud
High-Frequency, High-Severity	Common	Large	Priority control investment	(Rare - requires urgent action)
Low-Frequency, High-Severity	Rare	Large	Capital buffer + scenario analysis	Ransomware, major cyber incidents

Key Insight: Different risk profiles require different management strategies. High-frequency losses need process improvements and automation, while low-frequency/high-severity losses require capital buffers, insurance, and scenario analysis.

10.1.3.1 Modeling Challenges and Data Limitations

While sophisticated statistical models exist for operational risk (using Poisson distributions for frequency and lognormal distributions for severity), operational risk presents unique challenges:

- **Data scarcity**: Rare, severe events have few historical observations
- **Changing risk landscape**: New technologies create risks not present in historical data
- **Interdependencies**: Operational failures can trigger market, liquidity, and credit risks

These modeling challenges led regulators to develop the **Standardized Measurement Approach (SMA)**. Rather than requiring bank-specific LDA models under the Advanced Measurement Approach (AMA), SMA provides a practical, standardized framework using bank size and actual loss history to determine capital requirements.

10.2 Block B: Calculating Regulatory Capital under SMA

10.2.1 Theory: The Standardized Measurement Approach (SMA)

Regulators require banks to hold capital against potential operational losses. The **Standardized Measurement Approach (SMA)** is Basel's methodology for calculating this capital requirement. Introduced in December 2017 and implemented in January 2023, the SMA replaced earlier approaches (including the Advanced Measurement Approach, or AMA) with a single, consistent methodology applicable to all banks.

Conceptually, SMA can be viewed as a **regulatory proxy** for a high-confidence loss measure (similar to a 99.9% VaR/ES) that is easier to implement and compare across institutions.

10.2.1.1 Overview of the SMA Framework

The SMA combines two elements:

1. **Size-based capital**: The Business Indicator Component (BIC) scales with bank size and complexity
2. **Loss-based adjustment**: The Internal Loss Multiplier (ILM) adjusts capital based on actual loss experience

$$\text{Operational Risk Capital} = \text{BIC} \times \text{ILM}$$

Let's examine each component systematically.

10.2.1.2 Component 1: The Business Indicator (BI)

The **Business Indicator** measures bank size and operational complexity through three categories of financial activity:

$$\text{BI} = \text{ILDC} + \text{SC} + \text{FC}$$

where:

- ILDC ... Interest, Leases, Dividend Component
- SC ... Services Component
- FC ... Financial Component

$$\text{ILDC} = |\text{Interest Income} - \text{Interest Expense}| + \text{Dividend Income}$$

$$\text{SC} = \text{Fee Income} + \text{Other Service Income}$$

$$\text{FC} = |\text{Trading Gains/Losses}| + |\text{Investment Gains/Losses}|$$

Key principle: Absolute values (| |) ensure that losses don't artificially reduce the BI. A bank with large trading losses has high operational complexity and should not receive lower capital requirements.

Interpretation: The BI captures the scale of a bank's operations across lending, service provision, and trading activities. Larger BI indicates greater operational exposure.

10.2.1.3 Component 2: The Business Indicator Component (BIC)

The **Business Indicator Component** converts the BI into a baseline capital requirement using a **progressive three-tier structure**. BI is measured in **EUR billions**, and the formulas below produce BIC in **EUR billions** as well:

Small banks (BI $<$ €1 billion):

$$\text{BIC} = 0.12 \times \text{BI}$$

Medium banks (€1 billion BI $<$ €30 billion):

$$\text{BIC} = 0.12 + 0.15 \times (\text{BI} - 1)$$

Large banks (BI €30 billion):

$$\text{BIC} = 4.47 + 0.18 \times (\text{BI} - 30)$$

Understanding the coefficients:

- Small banks: 12% of BI
- Medium banks: €0.12B base + 15% marginal rate on amounts above €1 billion
- Large banks: €4.47B base + 18% marginal rate on amounts above €30 billion

Why progressive rates? Operational risk grows faster than bank size. Larger banks face greater organizational complexity, interconnected systems, and systemic importance. The increasing rates (12% $\rightarrow$ 15% $\rightarrow$ 18%) reflect this nonlinear risk relationship.

10.2.1.4 Component 3: Historical Loss Data

Banks must collect all operational loss events over a **rolling 10-year period**. This includes losses from all seven Basel categories, regardless of amount (though small losses below €20,000 may be excluded).

$$\text{Average Annual Loss} = \frac{\text{Sum of 10 years of losses}}{10}$$

This average captures the bank's actual loss experience, providing a backward-looking measure of operational risk profile.

10.2.1.5 Component 4: The Loss Component (LC)

The **Loss Component** scales historical losses to reflect the capital needed for extreme outcomes:

$$\text{LC} = 15 \times \text{Average Annual Loss (past 10 years)}$$

Why multiply by 15? The multiplier targets a high confidence level (roughly 99.9%) over a one-year horizon, derived from industry-wide loss distributions. The 10-year average captures "normal" losses; multiplying by 15 provides a buffer for rare but severe events. If a bank averaged €10M in annual losses, LC = €150M represents capital for a bad year at very high confidence.

10.2.1.6 Component 5: The Internal Loss Multiplier (ILM)

The **Internal Loss Multiplier** adjusts baseline capital (BIC) based on how the bank's loss experience compares to its size:

Small banks (BI $<$ €1 billion):

$$\text{ILM} = 1$$

Small banks don't receive adjustments—their capital equals BIC regardless of loss history.

Other banks (BI €1 billion):

$$\text{ILM} = \ln\left[e - 1 + \left(\frac{\text{LC}}{\text{BIC}}\right)^{0.8}\right]$$

where:

- e = Euler's constant (2.718)
- ln = Natural logarithm

Understanding ILM behavior:

Scenario	LC/BIC Ratio	ILM Value	Interpretation
Average performance	LC = BIC	ILM = 1.0	No adjustment to capital
Better than average	LC $<$ BIC	ILM $<$ 1.0	Capital reward (reduction)
Worse than average	LC $>$ BIC	ILM $>$ 1.0	Capital penalty (increase)
Perfect record	LC = 0	ILM 0.54	Maximum 46% capital reduction
Very poor record	LC » BIC	ILM » 1.0	Unbounded penalty

Why this formula? The logarithmic structure creates:

- **Diminishing returns**: Excellent records can't reduce capital below ILM 0.54 (46% reduction limit)
- **Escalating penalties**: Poor records face increasing penalties with no ceiling
- **Incentive alignment**: Improvements in loss performance yield tangible capital relief

10.2.1.7 Component 6: Final Capital Requirement

The SMA capital requirement combines BIC and ILM:

$$\text{Operational Risk Capital} = \text{BIC} \times \text{ILM}$$

Interpretation:

- Small banks cannot reduce capital below BIC, even with excellent loss history (ILM = 1)
- Medium and large banks can earn capital reductions (ILM < 1) or face penalties (ILM $>$ 1) based on performance
- This structure rewards institutions that invest in operational risk management

10.2.2 Activity 2: SMA Capital Calculation

Navigate to **Activity_2_SMA_Calculator** sheet in your Excel file.

You will calculate the operational risk capital requirement for **EuroBank**, a medium-sized European bank. The sheet provides a step-by-step template guiding you through all six steps of the SMA calculation.

Task 2.1: Calculate the **complete SMA capital requirement** for EuroBank.

Task 2.2 Interpret EuroBank's results:

- What is the operational risk capital as a percentage of BI?
- Is ILM greater than or less than 1.0? What does this tell you about EuroBank's historical loss performance compared to its size?

Task 2.3 Sensitivity analysis - What if EuroBank's historical losses had been twice as high?

- Recalculate LC, ILM, and final operational risk capital
- By how much would capital increase (in EUR billions and in %)?
- Is the increase in capital more or less than proportional to the increase in LC? What does this tell you about the nonlinearity of the ILM formula?
- What economic incentive does the ILM structure create for operational risk management?

10.3 Block C: Preventing Operational Risk Losses

While capital protects against losses, **prevention** is far more valuable. Let's explore how banks proactively manage operational risk.

10.3.1 Activity 3: Risk Controls and KRIs

Scenario: Your bank recently experienced an operational loss event:

> A trader in the commodities desk accumulated unauthorized positions over six months, resulting in a €42 million loss when market prices moved adversely. The trader concealed the positions by exploiting a gap in the trade confirmation system. The fraud was only discovered when the trader took sick leave and a colleague reviewed the portfolio.

Task 3.1 Which of Basel's seven categories does this loss fall under?

Task 3.2 Identify the root causes that enabled this loss to occur, selecting from different dimensions:

- **Process failures**: What controls were missing or inadequate?
- **System failures**: What technology gaps or weaknesses existed?
- **People failures**: What behavioral red flags or oversight gaps were missed?

Task 3.3 Your risk team has proposed two Key Risk Indicators (KRIs) to detect similar unauthorized trading:

KRI #1: Percentage of traders not taking mandatory 10 consecutive days leave annually *Threshold*: $> 15\%$ triggers investigation

KRI #2: Number of manual trade confirmation overrides per trader per month *Threshold*: > 5 overrides triggers investigation

For each KRI:

- Evaluate effectiveness of each KRI: Would it have detected this fraud? Why or why not?
- Design **one additional KRI** that addresses a gap not covered by KRIs #1 and #2. Specify the metric, threshold, and rationale.

Task 3.4 Your risk committee proposes three responses to prevent future incidents:

1. **Invest in prevention**: Upgrade trade monitoring system (€5M upfront cost)
2. **Accept and insure**: Purchase operational risk insurance with €10M deductible (€2M annual premium)
3. **Hold capital only**: Allocate SMA capital but no additional controls

Briefly discuss: When would each approach be most appropriate?

10.3.2 Theory Block C: Risk Control and Self-Assessment (RCSA)

Risk Control and Self-Assessment (RCSA) is a structured process where business units:

1. **Identify risks**: Map processes and identify failure points
2. **Assess impact**: Estimate potential loss frequency and severity
3. **Evaluate controls**: Review existing safeguards
4. **Close gaps**: Implement additional controls where needed
5. **Monitor**: Track effectiveness over time

Key Risk Indicators (KRIs) serve as **early warning signals** for emerging operational risks. Effective KRIs have these characteristics:

- **Forward-looking**: Predict problems before they materialize
- **Measurable**: Objective, quantifiable metrics
- **Actionable**: Clear threshold triggers investigation
- **Relevant**: Directly linked to specific risk types

Common KRI Examples:

Risk Type	KRI Example	Warning Threshold
Internal Fraud	% employees not taking 10+ consecutive leave days	$> 15\%$
System Failures	System downtime hours per month	> 8 hours
Process Errors	Failed transaction rate	$> 0.5\%$
Cyber Risk	Phishing click rate in training	$> 10\%$
Turnover Risk	Staff turnover in critical functions	$> 20\%$ annually
Oversight Risk	Supervisor-to-staff ratio	$< 1{:}12$

Key Insight

Unlike credit or market risk, operational risk cannot be diversified away or hedged. It must be **prevented, monitored, and capitalized**. This makes operational risk management fundamentally about **organizational culture, processes, and systems**—not just quantitative models.

10.4 Key Takeaways

1. **Operational risk is heterogeneous**: Seven distinct Basel categories show different frequency/severity patterns, from high-frequency processing errors to low-frequency cyber incidents, each requiring tailored management approaches.
2. **Heavy-tailed distributions**: Most operational losses are small, but rare extreme losses (like ransomware attacks) drive capital requirements. This pattern challenges traditional statistical modeling approaches.
3. **SMA balances size and experience**: The Standardized Measurement Approach sets capital based on bank size (BIC) adjusted for actual loss history (ILM). Banks with better operational risk management earn lower capital requirements through ILM < 1.
4. **Prevention is more valuable than capital**: While SMA capital provides a buffer, proactive controls and early warning systems (KRIs) prevent losses from occurring. A €5M investment in trade monitoring systems can prevent a €42M fraud.
5. **Operational risk management is organizational, not just quantitative**: Unlike market or credit risk, operational risk cannot be hedged or diversified away. Effective management requires strong processes, system controls, employee oversight, and organizational culture—not just mathematical models.

11 Bank Regulation and Economic Capital

References

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 25 - Basel I, Basel II, and Solvency II
  - Chapter 26 - Basel II.5, Basel III, and Other Post-Crisis Changes
  - Chapter 27 - Fundamental Review of the Trading Book
  - Chapter 28 - Economic Capital and RAROC

Learning Outcomes:

- Explain and quantify economic capital, including its link to target ratings and aggregation across major risk types.
- Calculate and interpret RAROC for business units, and use it to support capital allocation and strategic decisions.
- Trace the evolution of bank capital regulation from Basel I to Basel III and FRTB, focusing on intuition, strengths, weaknesses, and unintended consequences.
- Explain the concepts of G-SIBs, D-SIBs, SIFIs, TLAC, and CoCos, and discuss their roles in addressing "too big to fail" and loss absorption.

11.1 Block A: Economic Capital and RAROC

11.1.1 Theory: Beyond Regulatory Minimum

11.1.1.1 Economic Capital

- **Economic capital** is the bank's own, model-based estimate of how much capital it needs to cover unexpected losses over one year at a chosen confidence level.
- **Regulatory capital** is the minimum capital required by supervisors under Basel rules, based on standardized or approved internal models.

Banks often link the confidence level for economic capital to a target external rating:

Rating	Default Probability	Implication for economic capital
AAA	0.01%	Very high capital: losses covered in almost all scenarios
AA	0.02%	High capital: can absorb very severe losses
A	0.05%	Solid capital: can absorb severe but less extreme losses
BBB	0.20%	Moderate capital: can absorb typical stress scenarios

Choosing a higher target rating (e.g. AAA instead of BBB) means the bank wants default to be more unlikely, so it must hold more economic capital to make sure the modeled 1-year loss exceeds capital only in very rare states of the world.

In practice you compare **available capital** with two benchmarks:

- **Regulatory capital requirement** (Basel): external minimum you must satisfy
- **Economic capital requirement** (internal): capital needed to support the target rating

The binding constraint is the larger of the two. If available capital just meets the regulatory minimum but is **below** the economic capital requirement, the bank is economically undercapitalized even though it is formally compliant with regulation.

Economic capital aggregates all main risk types from previous weeks:

- **Market risk** (Weeks 5–6): one-year VaR/ES based on trading and ALM positions
- **Liquidity risk** (Week 8): cost of funding or forced liquidation in stress
- **Credit risk** (Week 9): default and migration risk on loans and derivatives
- **Operational risk** (Week 10): loss distribution for fraud, systems, legal, etc.
- **Business / strategic risk**: model, reputational and business-model risk

11.1.1.2 Mathematical View: Economic Capital as a Risk Measure

Let L denote the random loss of the bank over one year (positive = loss). Denote by $\mathbb{E}[L]$ the **expected loss** (EL), which should be covered by pricing and provisions, and by $\rho(L)$ a **risk measure** of the loss distribution capturing unexpected losses.

A general way to define **economic capital** is

$$\mathrm{EC} = \rho(L) - \mathbb{E}[L],$$

so that EC is the extra capital buffer, on top of EL, needed to protect the bank against rare but severe losses. Two common choices for $\rho(\cdot)$ are Value-at-Risk and Expected Shortfall.

Link to target rating: if the bank targets a one-year default probability p^* (e.g. $p^* = 0.03\%$ for an A rating), it typically chooses $\alpha = 1 - p^*$. Then, using either VaR or ES as $\rho(\cdot)$, the economic capital level EC is calibrated so that the modeled probability that losses exceed EL + EC is roughly p^*, matching the desired external rating.

11.1.1.3 Aggregating Economic Capital

If we simply add economic capital across risks

$$E_{\text{total}} = \sum_{i=1}^{n} E_i,$$

we implicitly assume perfect correlation across risk types and business units. This is usually too conservative, so banks use a simple correlation-based aggregation:

$$E_{\text{total}} = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} E_i E_j \rho_{ij}},$$

where ρ_{ij} is the (low to moderate) correlation between risks i and j. This produces a lower, diversified total EC.

The simple sum therefore overstates required capital, while the fixed correlation-matrix approach can understate it, because correlations between risks tend to jump toward 1 in crises and losses become more tightly linked in the tails. In practice, banks try to correct for this by using stress scenarios and portfolio models (sometimes with copulas or stressed correlations) to capture higher dependence in bad states of the world when sizing economic capital.

11.1.1.4 RAROC: Risk-Adjusted Return on Capital

RAROC measures economic profitability per unit of economic capital:

$$\text{RAROC} = \frac{\text{Revenues} - \text{Costs} - \text{Expected Losses}}{\text{Economic Capital}}$$

- **Numerator**: risk-adjusted profit (after expected losses)
- **Denominator**: economic capital allocated to the business unit

Interpretation:

- RAROC $>$ cost of equity $\rightarrow$ value creation
- RAROC $<$ cost of equity $\rightarrow$ value destruction

RAROC lets you compare very different businesses (retail, trading, asset management) on a common, risk-adjusted basis and decide where to grow or shrink.

11.1.2 Activity 1: Business Unit RAROC Analysis and Strategic Decisions

Download Economic Capital Data

Task 1.1 Lets start with sheet `EC_RAROC`. For each business unit, calculate: Profit before expected losses, Risk-adjusted profit, and RAROC. Interpret results. What is total firm RAROC?

Task 1.2 Using the correlation matrix, compute diversified total economic capital. Compare this to the simple sum of stand-alone EC. What is the diversification benefit in $ and in %?

Task 1.3 Solve credit portfolio optimization to maximize RAROC under economic capital constraint in excel sheet `EC_Optim`. A bank has $100 million to allocate across four loan categories. Total economic capital available is $10 million. What is the optimal allocation? How does the result change if total economic capital is constrained to $8 million?

Task 1.4 Discussion Points:

- Why use economic capital instead of regulatory capital for internal decisions?
- What are the limitations of RAROC as a performance measure?
- How should a bank handle the time horizon mismatch between short-term performance measurement and long-term capital requirements?

11.2 Block B: The Evolution of Bank Regulation

11.2.1 Big Picture: Why Regulation Keeps Changing

Bank regulation tries to balance two goals that are in tension:

- Keep banks safe enough to avoid costly crises and taxpayer bailouts
- Allow banks to provide credit, liquidity, and risk transfer to the real economy

Before 1988, each country had its own simple capital-to-asset rules. These were easy to apply but did a poor job of capturing the risks from international lending, derivatives, and securitization. Large banks could also engage in "regulatory shopping" by booking business in countries with weaker rules.

The Basel Committee's frameworks (Basel I, II, II.5, III and now FRTB) can be read as a sequence of **patches** to visible weaknesses:

- Basel I introduces a simple, global standard but is too crude.
- Basel II makes capital more risk-sensitive but relies heavily on banks' own models.
- The Global Financial Crisis reveals that both capital levels and risk measures were inadequate, especially in trading books and off-balance-sheet exposures.
- Basel II.5 and Basel III push for **more and better-quality capital**, add liquidity and leverage constraints, and try to address model risk and procyclicality.
- FRTB focuses specifically on **tail risk** and **liquidity risk** in the trading book.

11.2.2 Basel I–III and FRTB in One Narrative

- **Basel I (1988)** – "simple buckets"
  - Introduces **risk-weighted assets (RWA)** with a small number of risk weights (0%, 20%, 50%, 100%) and an 8% minimum capital ratio.
  - Strength: first common international standard; recognizes that not all assets are equally risky.
  - Weakness: the buckets are **very crude** (AAA = B-rated corporate), and initially only credit risk is covered.
  - **Unintended consequence:** encourages **regulatory arbitrage** through securitization and balance-sheet engineering (move "good" assets off balance sheet, keep risky exposures, but report lower RWA).
- **Basel II (around 2007)** – "more risk-sensitive, more complex"
  - Introduces internal ratings-based (IRB) approaches and an explicit **Three Pillar** structure:
    - Pillar 1: minimum capital
    - Pillar 2: supervisory review
    - Pillar 3: market discipline via disclosure
  - Strength: better differentiation of credit risk, inclusion of **operational risk**, more alignment with banks' own models.
  - Weakness: heavy reliance on model assumptions and historical data; capital tends to be lowest in good times and highest in bad times (**procyclicality**).
  - **Unintended consequence:** complex rules and model freedom create scope for underestimating tail risk and for "optimizing" models to reduce capital.

- **Global Financial Crisis (2007–2009)** – "stress test of the system"
  - Trading books, securitizations, and off-balance-sheet vehicles produce losses far beyond what models predicted.
  - Market liquidity evaporates; positions that looked "safe" under normal assumptions become impossible to unwind.
  - Many banks meet formal capital requirements right before the crisis, but are clearly undercapitalized once stress hits.
- **Basel II.5 and Basel III (2010–2023)** – "more and better-quality capital, plus new constraints"
  - **Basel II.5**: adds stressed VaR and an incremental risk charge to force trading book models to reflect crisis conditions.
  - **Basel III**:
    - Raises the level and quality of capital (**CET1**, capital conservation and countercyclical buffers).
    - Introduces a non-risk-weighted **leverage ratio** as a backstop.
    - Adds liquidity ratios (**LCR**, **NSFR**) to address funding and market liquidity risk.
  - Strength: banks enter the next crisis with more and better-quality capital, less reliance on short-term wholesale funding.
  - Weakness: still complex, still model-based; some risks (e.g. cyber, conduct, climate, business-model risk) remain only partially captured.
  - **Unintended consequences:**
    - Some activities move into the less regulated "shadow banking" sector.
    - Banks change their asset mix in response to risk weights and leverage constraints.
- **FRTB (Fundamental Review of the Trading Book)** – "tail risk and liquidity"
  - Replaces VaR with **Expected Shortfall (ES)** at 97.5% for regulatory capital.
  - Introduces **liquidity horizons**: capital charges are higher for risk factors that are slow to hedge or unwind in stress.
  - Tightens the boundary between **trading book** and **banking book** to reduce arbitrage between the two.
  - Goal: make capital more responsive to **extreme moves and illiquidity**, not just day-to-day volatility.

11.2.3 Activity 2: Regulation and Systemic Risk

1. "When a steel company goes bankrupt, other companies in the same industry benefit because they have one less competitor. But when a bank goes bankrupt other banks do not necessarily benefit." Explain this statement.
2. "The existence of deposit insurance makes it particularly important for there to be regulations on the amount of capital banks hold." Explain this statement.
3. Under Basel I and II, banks sometimes used **regulatory arbitrage** to reduce their capital without reducing underlying risk (for example, via securitization or moving positions between trading and banking books). Give an example of such behavior and discuss why it is problematic for financial stability.

4. What is the difference between the **trading book** and the **banking book**? Why are regulators concerned about specifying whether an instrument should be one or the other in the FRTB and Basel frameworks?

5. A bank can be compliant with all Basel III ratios and still be risky. Which risks remain imperfectly captured by regulatory capital and leverage requirements? How should supervisors and market participants respond?

11.3 Block C: Too Big to Fail: G-SIBs, D-SIBs, TLAC and CoCos

Even with better rules, some institutions are so large and interconnected that their failure threatens the system as a whole. This is the **"too big to fail"** problem.

Systemically important banks

- **Global Systemically Important Banks (G-SIBs)** are very large, internationally active banks whose failure would have global repercussions.
- **Domestic Systemically Important Banks (D-SIBs)** are critical for a particular national banking system, even if they are not global players.
- More broadly, regulators talk about **Systemically Important Financial Institutions (SIFIs)**, which can include large insurers and other non-banks.

These institutions face **capital surcharges** on top of Basel III minimums (for G-SIBs, roughly +1–3.5% CET1, depending on their systemic importance bucket). The idea is:

- They benefit from an implicit "too big to fail" funding advantage.
- In return, they must hold more **loss-absorbing capacity** and prepare credible resolution plans.

For G-SIBs, regulators also require **Total Loss-Absorbing Capacity (TLAC)**:

- TLAC consists of equity plus certain long-term debt that can be written down or converted into equity in resolution.
- The goal is to enable a **"bail-in"**: shareholders and eligible creditors absorb losses, rather than taxpayers, while the essential parts of the bank continue to operate.

Contingent convertible bonds (CoCos) are a prominent example of bail-in style instruments:

- They are bonds that convert into equity or are written down when a trigger is hit (for example, when the bank's CET1 ratio falls below a threshold, or when regulators decide the bank is no longer viable).
- Investors receive higher coupons to compensate for the risk of conversion or write-down.
- If they meet certain criteria, they count as **Additional Tier 1 (AT1)** capital.

From different perspectives:

- **Banks** like CoCos because they are cheaper than equity in normal times but help strengthen capital in stress.
- **Regulators** like them because they can provide **automatic recapitalization** and reduce the need for sudden dilutive equity issues or taxpayer support.
- **Investors** bear the downside: in a severe stress, they can lose some or all of their investment.

11.3.1 Activity 3: Discussion on Systemic Importance and CoCos

1. Explain how **CoCo bonds** work. Why are they attractive to (a) banks and (b) regulators? What are the main risks from the point of view of (c) investors and (d) financial stability?
2. Do you think higher capital requirements and TLAC-style loss-absorbing debt for systemically important institutions will really solve the "too big to fail" problem, or will risk migrate elsewhere in the system? Support your answer with arguments from this and previous weeks.

11.4 Key Takeaways

- **Economic capital and RAROC** provide an internal view of risk and performance: they aggregate market, credit, liquidity, operational, and business risks and allow banks to compare business units on a common, risk-adjusted basis.
- **Regulatory capital frameworks** (Basel I $\rightarrow$ Basel II $\rightarrow$ Basel III/FRTB) are best seen as a series of "patches" to visible weaknesses and arbitrage, improving risk sensitivity and resilience but also increasing complexity and creating new incentives.
- **Systemically important institutions** (G-SIBs, D-SIBs, SIFIs) and tools like **TLAC** and **CoCos** are designed to make large bank failures more orderly by shifting losses from taxpayers to shareholders and creditors, raising questions about who ultimately bears risk.
- For both supervisors and bank management, the central challenge is to align **regulation, internal capital models, and incentives** ex ante so that banks remain resilient in stress without unduly restricting their ability to provide useful financial intermediation.

12 Test 2

13 Final Exam